ESE 4007 Complete Study Pack

Smart Grid Technology · KUET · past papers 2020, 2022, 2023 · answers from standard textbooks (no course slides available)

24-hour priority list

You answer **3 of 4 questions in each part**, so each part has its own list. Study down both lists until the green **pass line**: about 4 h for Part A and 4 h for Part B, which leaves time for sleep and a last read of the one-shot notes. Alternate between parts (A block 1, B block 1, A block 2 …) so neither is left short. **Chance**: how often the topic came up, newer papers weigh more (2023 ×3, 2022 ×2, 2020 ×1). **Covered**: share of that part's past-paper marks you have studied once you finish the row.

Next if you have time: B4 (swing equation and equal area) is the one hard topic that keeps coming back; the power-angle graph makes it quick to understand. B5 (2020 numerical) appeared only once, in the oldest paper, so learn it last.

Section A: Smart grid concepts, distribution, communication, metering, PMU, load flow and analysis tools

Block 1: very likely and easy · 2.9 h

Topic	Chance	Level	Covered
A1 What a smart grid is: definition, conventional grid vs smart grid, architecture and components, key drivers, stakeholders, need for Bangladesh	Very likely 100%	Easy	12%
A3 Enabling technologies: information and communication; sensing, measurement, control and automation; power electronics and energy storage	Very likely 100%	Easy	24%
A6 Smart meter, AMR vs AMI, how AMI works, architecture for communication between smart meter and utility server	Very likely 100%	Easy	34%
A9 Load flow in smart grids: classical methods, old vs desired load flow, essential features, why needed and challenges	Very likely 100%	Easy	42%
A4 Distributed generation (DG) and microgrid: definition, advantages, configuration, grid-connected and islanded operation	Very likely 83%	Easy	53%
A7 Demand side management (DSM): load shifting, valley filling, peak clipping (load-shape curves)	Very likely 83%	Easy	60%
A2 Active vs passive distribution network; active distribution management scheme (block diagram)	Very likely 83%	Easy	67%

Block 2: very likely, medium effort · 3.0 h

Topic	Chance	Level	Covered
A5 Smart grid communication: SCADA, HAN/NAN/WAN, and where Bluetooth, ZigBee, WLAN, WPAN and WiMAX are used	Very likely 100%	Medium	78%
✓ Pass line: 78% of Section A marks covered, enough to choose 3 of the 4 questions (about 3.7 h)			
A10 Performance analysis tools: contingency analysis, security-constrained OPF, DSOPF, static security assessment (SSA), congestion management; GIS	Very likely 100%	Medium	89%
A8 Phasor measurement unit (PMU) and synchronised phasor measuring system (WAMS)	Very likely 83%	Medium	95%
A11 Multi-agent systems (MAS): IDAPS and decentralised meter reading	Very likely 83%	Medium	100%

Section B: Power system control and stability, power quality, intelligence, security, renewables

Block 1: very likely and easy · 2.5 h

Topic	Chance	Level	Covered
B8 Computational intelligence and decision tools: ANN, expert systems, AHP, decision support tools, stability tools and computational challenges	Very likely 100%	Easy	18%
B9 Interoperability and cyber security: threats, false data injection attacks, mitigation	Very likely 100%	Easy	30%
B3 Reactive power and voltage control: why the grid needs reactive power, control methods, AVR, SVC, improving P_{max}	Very likely 100%	Easy	40%
B11 Barriers and solutions, standards and research areas, the simplest and the most complex thing about the smart grid	Very likely 100%	Easy	48%
B10 Renewables and the smart grid: integration challenges, education and training, V2G, sustainable energy options, efficiency and DG control	Very likely 83%	Easy	57%
B7 Power quality: issues in a smart grid, importance of power-quality management and audit	Very likely 83%	Easy	61%

Block 2: very likely, medium effort · 2.2 h

Topic	Chance	Level	Covered

B2 Frequency control: why grid frequency matters, turbine speed governing system, load frequency control loop, AGC	Very likely 100%	Medium	72%
B6 Voltage stability: why instability occurs, preventive control, assessment techniques (P-V, Q-V, VIPI, indices)	Very likely 100%	Medium	83%
✔ Pass line: 83% of Section B marks covered, enough to choose 3 of the 4 questions (about 4.0 h)			
B1 Buses and bus types, infinite bus, bus admittance matrix Y_{bus} and proof of $I_{bus} = Y_{bus}V_{bus}$	Very likely 83%	Medium	88%

Block 3: very likely numericals · 1.0 h

Topic	Chance	Level	Covered
B4 Rotor angle stability: steady-state vs transient, swing equation, small-disturbance solution $\omega_n = \sqrt{(\pi f_0 P_s/H)}$, equal area criterion	Very likely 83%	Hard	98%

Only if time is left · 1.0 h

Topic	Chance	Level	Covered
B5 Numerical: voltage behind transient reactance and transient power-angle equation (with and without saliency)	Possible 17%	Hard	100%

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★ One-shot notes

One-shot notes: ESE 4007 Smart Grid Technology

Smart grid in one line

Old grid + sensors (eyes) + two-way communication (nerves) + automatic control (brain). Digital, two-way power and data, distributed generation, self-healing, active customers.

Drivers: demand, ageing assets, reliability, renewables, climate, efficiency, customer choice, ICT. Bangladesh: losses/theft, peak gap, blackouts, renewables target, costly imported fuel.

Stakeholders

Generators, transmission/system operator (PGCB, NLDC), distribution utilities (DPDC, DESCO, BREB), consumers/prosumers, regulator (BERC, SREDA), vendors/ICT, aggregators, researchers/standards bodies.

Active vs passive distribution

Passive: no DG, one-way, fit and forget, simple protection. Active: DG/storage/flexible loads, two-way, real-time control.

Loop: measure → communicate → state estimate + forecast → control (Volt/VAR, DG dispatch, reconfiguration, DR, FLISR) → act.

Three technology families

ICT: fibre, PLC, ZigBee, Wi-Fi, WiMAX, cellular; protocols IEC 61850, DNP3. Sensing/control: smart meters, PMU, sensors, IEDs, SCADA/DMS, automation. Power electronics: inverters, FACTS, HVDC; storage: batteries, pumped hydro, flywheel, supercapacitor, SMES, CAES.

DG and microgrid

DG = small (kW-50 MW) generation in the distribution network near loads. Advantages: lower losses, better voltage, reliability, deferred investment, renewables, CHP, quick.

Microgrid: DGs + storage + loads + MGCC + static switch at PCC. On-grid: grid sets V, f; DGs in PQ mode. Islanded: switch opens, V/f or droop control, shed non-critical loads, resynchronise later.

Communication

SCADA: field devices/RTU/IED → network → master station + HMI. Advantages: real-time monitoring, remote control, faster restoration, efficiency, data, safety.

HAN (home, 10-100 m: ZigBee, Bluetooth, Wi-Fi) → NAN (neighbourhood, 1-10 km: RF mesh, PLC, WiMAX) → WAN (backbone: fibre, WiMAX, LTE).

Smart meter and AMI

Meter → NAN → concentrator → WAN → head-end → MDMS → billing, OMS, DMS, portal.

AMR one-way billing reads; AMI two-way: interval data, remote connect/disconnect, ToU and prepaid, outage alerts, HAN link.

DSM

Change timing/amount of use to flatten the load curve. Peak clipping (cut peak: direct load control), valley filling (add off-peak load: night EV charging), load shifting (move peak to off-peak: ToU, thermal storage). Plus conservation, load growth, flexible load.

PMU and WAMS

Measures V, I phasors + frequency + ROCOF, GPS time-stamped (1 µs), 25-60 per second. Blocks: CT/PT inputs, anti-aliasing filter, A/D, GPS + phase-locked oscillator, DFT microprocessor, modem.

WAMS: PMUs → local PDC → super PDC → monitoring, state estimation, protection.

Load flow

Buses: slack (V, δ), PV (P, $|V|$), PQ (P, Q). Methods: Gauss-Seidel (simple, slow), Newton-Raphson (fast convergence), fast decoupled (fastest), DC (linear).

Smart grid needs: stochastic, three-phase unbalanced, DG and two-way flow, high R/X robust, real-time with PMU/AMI data, large-scale.

Analysis tools

Contingency = what if one element trips (N-1). SSA = steady post-contingency limits check: secure / alert / emergency. SCOPF = cheapest dispatch still secure after contingencies. DSOPF = OPF + time + uncertainty (adaptive dynamic programming). Congestion = line at its limit: re-dispatch, FACTS, pricing. GIS: asset maps, outages, crews, planning.

Multi-agent systems

Agent: autonomous, reactive, pro-active, social. IDAPS agents: control (island/reconnect), DER, user (critical loads), database. Meter MAS: meter agents → collector agents → billing, database, monitoring agents.

Y_{bus}

Y_{ii} = sum of admittances at bus i (including shunt); $Y_{ij} = -y_{ij}$. KCL at each bus ⇒ $I_i = \Sigma Y_{ij}V_j$ ⇒ $I_{bus} = Y_{bus}V_{bus}$. Symmetric, sparse.

Infinite bus: constant V and f whatever the power.

Frequency control

Frequency = generation-load balance. Governor chain: fly-ball → linkage → hydraulic amplifier → valve; droop R leaves $\Delta f = -R\Delta P_L/(1 + DR)$.

LFC loop: governor $1/(1 + sT_g)$ → turbine $1/(1 + sT_t)$ → $1/(2Hs + D)$, feedback 1/R. AGC: integral of ACE = $\Delta P_{tie} + B\Delta f$; restores 50 Hz and tie flows; essential with renewables.

Reactive power and voltage

P ↔ frequency, Q ↔ voltage. Q devices: AVR/excitation, shunt capacitors/reactors, series capacitors, synchronous condensers, OLTC, SVC, STATCOM, smart inverters.

AVR: sensor → compare with V_{ref} → amplifier → exciter → field. SVC = TCR + TSC + filters. P_{max} = EV/X: raise E, V; cut X.

Rotor-angle stability

Steady-state = small slow changes; transient = large sudden faults. Swing: $(H/\pi f_0)d^2\delta/dt^2 = P_m - P_{max}\sin\delta$.

Small disturbance: $P_s = P_{max}\cos\delta_0$, $s = \pm j\sqrt{(\pi f_0 P_s/H)}$ ⇒ $\omega_n = \sqrt{(\pi f_0 P_s/H)}$; $P_s < 0$ unstable.

EAC: stable if A_2 (decelerating) can equal A_1 (accelerating) before $\delta_{max} = 180° - \delta_0$. $\cos\delta_{cr} = (\pi - 2\delta_0)\sin\delta_0 - \cos\delta_0$.

Machine numerical (2020)

$I = 0.625\angle -36.87°$. No saliency: $E' = 1 + j0.3I = 1.1226\angle 7.68°$, $P = 3.742 \sin\delta$.

Saliency: $E_Q = 1 + j0.6I = 1.2612\angle 13.76°$; $I_d = 0.4832$; $E'_q = 1.1162$; $P = 3.721 \sin\delta - 0.833 \sin 2\delta$.

Voltage stability

Causes: heavy load, long lines, Q shortage/generator limits, outages, OLTC action, motor loads. Prevent: Q reserves, capacitors/SVC, generator V set-points, block OLTC, re-dispatch, load shedding, PMU monitoring.

Techniques: P-V, Q-V, CPF, modal analysis, indices (L-index, VIPI), dynamic simulation, PMU/AI. VIPI = angle between load vector and the midpoint of the two load-flow solutions → 0 at collapse.

Power quality

Sag, swell, interruption, transients, harmonics, flicker, unbalance, frequency deviation. Matters for sensitive loads, heating/losses, inverter-rich grid. PQ audit: measure, find source, fix, comply.

Intelligence tools

ANN: layers of weighted neurons trained on data; forecasting, security assessment, fault detection, theft detection. ES: knowledge base + inference engine; alarm processing, fault diagnosis, restoration.

AHP: hierarchy → pairwise 1-9 → weights (eigenvector) → CR < 0.1 → synthesis.
Computational challenges: size, uncertainty, real-time, non-linear optimisation, multi-time-scale, data/cyber, models.

Cyber security

Interoperability = exchange and use information (IEC 61850, CIM, DNP3, IEEE 2030). Threats: malware, DoS, man-in-the-middle, meter tampering, FDI, GPS spoofing; huge attack surface.

FDI: a = Hc passes the residual test → wrong state, wrong dispatch, overloads. Mitigation: defence in depth, authentication, encryption, protected key meters, PMUs, anomaly detection, response plans, training.

Renewables, V2G, barriers

Renewable challenges: variable, uncertain, low inertia, distributed (forecasting, storage, protection, PQ, markets). V2G: bidirectional charger, charge off-peak, discharge at peak, frequency regulation; battery wear.

Barriers: cost, regulation, standards, security, skills, consumer acceptance → phased rollout, policy, standards, training, awareness.

Abbreviations used in this pack

Every short form, written out. The answers also spell each one out the first time it appears.

Short form	Full meaning
A/D	analog-to-digital
ACE	area control error
ACL	agent communication language
ADB	Asian Development Bank
ADMS	advanced distribution management system
AGC	automatic generation control
AHP	analytical hierarchy process (the paper calls it analytical hierarchical programming)
AI	artificial intelligence
AMI	advanced metering infrastructure
AMR	automatic meter reading
ANN	artificial neural network
ANSI	American National Standards Institute
AVR	automatic voltage regulator
BAN	building area network
BERC	Bangladesh Energy Regulatory Commission
BREB	Bangladesh Rural Electrification Board
CAES	compressed air energy storage
CHP	combined heat and power
CI	consistency index
CIA	confidentiality, integrity, availability
CIM	common information model
CIP	critical infrastructure protection
CIS	customer information system
CO_2	carbon dioxide
COSEM	Companion Specification for Energy Metering
CPF	continuation power flow
CR	consistency ratio
CT	current transformer
DER	distributed energy resource
DESCO	Dhaka Electric Supply Company
DFT	discrete Fourier transform
DG	distributed generation
DLMS	Device Language Message Specification
DLR	dynamic line rating
DMS	distribution management system
DMZ	demilitarised zone (buffer network)

DNP3	Distributed Network Protocol 3
DoS	denial of service
DPDC	Dhaka Power Distribution Company
DR	demand response
DSL	digital subscriber line
DSM	demand side management
DSOPF	dynamic stochastic optimal power flow
DST	decision support tool
DVR	dynamic voltage restorer
EAC	equal area criterion
EMS	energy management system
ES	expert system
EV	electric vehicle
FACTS	flexible AC transmission systems
FAN	field area network
FDI	false data injection
FDLF	fast decoupled load flow
FIPA	Foundation for Intelligent Physical Agents
FLISR	fault location, isolation and service restoration
G2V	grid-to-vehicle
GIS	geographic information system
GPS	Global Positioning System
GS	Gauss-Seidel
HAN	home area network
HES	head-end system
HMI	human-machine interface
HVDC	high-voltage direct current
Hz	hertz (cycles per second)
ICT	information and communication technology
IDAPS	intelligent distributed autonomous power system
IEC	International Electrotechnical Commission
IED	intelligent electronic device
IEEE	Institute of Electrical and Electronics Engineers
IoT	Internet of Things
IPP	independent power producer
IT	information technology
JADE	Java Agent DEvelopment framework

KCL	Kirchhoff's current law
KVL	Kirchhoff's voltage law
kW	kilowatt
kWh	kilowatt-hour
LED	light-emitting diode
LFC	load frequency control
LNG	liquefied natural gas
LTE	Long Term Evolution (4G cellular)
MAS	multi-agent system
MDMS	meter data management system
MGCC	microgrid central controller
MTU	master terminal unit (master station)
MVA	mega volt-ampere
MW	megawatt
NAN	neighbourhood area network
NERC	North American Electric Reliability Corporation
NESCO	Northern Electricity Supply Company
NIST	National Institute of Standards and Technology (USA)
NLDC	National Load Despatch Centre
NR	Newton-Raphson
OLTC	on-load tap changer
OMS	outage management system
OpenADR	Open Automated Demand Response
OPF	optimal power flow
OT	operational technology
PCC	point of common coupling
PDC	phasor data concentrator
pf	power factor
PGCB	Power Grid Company of Bangladesh (now Power Grid Bangladesh PLC)
PLC	programmable logic controller (or power line communication, by context)
PMU	phasor measurement unit
PPS	pulse per second
PQ	power quality (also PQ bus = P and Q specified)
PSS	power system stabiliser
PT	potential (voltage) transformer
pu	per unit (value / base value)
PV	photovoltaic (solar cells); also PV bus = P and $\|V\|$ specified

RF	radio frequency
RI	random index
ROCOF	rate of change of frequency
RTU	remote terminal unit
SCADA	supervisory control and data acquisition
SCOPF	security-constrained optimal power flow
SG	smart grid
SMES	superconducting magnetic energy storage
SMPS	switched-mode power supply
SREDA	Sustainable and Renewable Energy Development Authority
SSA	static security assessment
STATCOM	static synchronous compensator
SVC	static var compensator
TCR	thyristor-controlled reactor
TCSC	thyristor-controlled series capacitor
THD	total harmonic distortion
Tk	Taka
TLS	Transport Layer Security
ToU	time of use
TSC	thyristor-switched capacitor
UPFC	unified power flow controller
UPS	uninterruptible power supply
UTC	Coordinated Universal Time
V2G	vehicle-to-grid
V2H	vehicle-to-home
VEE	validation, editing and estimation
VFD	variable frequency drive
VIPI	voltage instability proximity index
VPN	virtual private network
WAMPAC	wide area monitoring, protection and control
WAMS	wide area monitoring system
WAN	wide area network
WiMAX	Worldwide Interoperability for Microwave Access (IEEE 802.16)
WLAN	wireless local area network (Wi-Fi)
WPAN	wireless personal area network
WZPDCL	West Zone Power Distribution Company Ltd

Section A: Smart grid concepts, distribution, communication, metering, PMU, load flow and analysis tools

A1 What a smart grid is: definition, conventional grid vs smart grid, architecture and components, key drivers, stakeholders, need for Bangladesh Must study Very likely · 100% Easy

PAST-PAPER QUESTIONS

- **2023 Q1(a)** [10] Make a comparison between the conventional power system and a smart grid.
- **2022 Q1(b)** [10] Who are the main stakeholders of smart grid systems? Describe their functions.
- **2022 Q1(c)** [10] Mention the key drivers of smart grid. Is a smart grid necessary for Bangladesh? Justify your opinion.
- **2020 Q1(a)** [3] What is a smart grid?
- **2020 Q1(b)** [12] Illustrate the architecture of the smart grid design. What are the main functions of smart grid components?

★ REMEMBER IT

- Smart grid = old grid + **eyes** (sensors, smart meters, PMUs) + **nerves** (two-way communication) + **brain** (automatic control, software). Old grid: one-way, central, blind, manual. Smart grid: two-way, distributed, observable, self-healing.
- Stakeholders: generators, transmission and distribution utilities, consumers/prosumers, regulators, equipment vendors, ICT (information and communication technology) providers.

Before you start: what makes a grid "smart"

The old grid is like a one-way water pipe system: big power stations push electricity through transmission lines and distribution feeders to homes, and the utility learns about a fault only when customers phone in.

A **smart grid** adds a **nervous system** to that pipe system: sensors and smart meters everywhere (eyes), two-way digital communication (nerves), and computers that analyse and control automatically (brain). Now power and information both flow both ways, homes can also generate (rooftop solar), and the grid can heal itself.

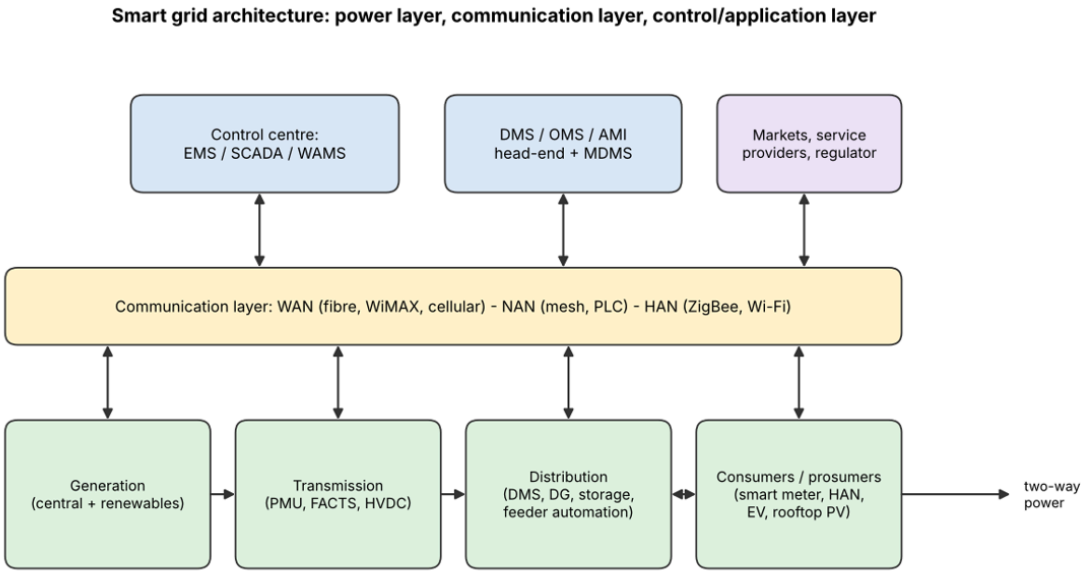
Definition (2020 1a)

A **smart grid** is an electricity network that uses digital and other advanced technologies - two-way communication, sensing and measurement, automatic control and computer intelligence - to monitor and manage the transport of electricity from all generation sources to meet the varying demands of end users, so that the grid is more reliable, efficient, secure, economical and environment-friendly. (Momoh: "the integration of power, communication and information technologies for an improved electric power infrastructure serving loads while providing an ongoing evolution of end-use applications".)

Conventional power system vs smart grid (2023 1a)

Feature	Conventional grid	Smart grid
Technology	Electromechanical	Digital
Communication	One-way (or none)	Two-way, real time
Power flow	One-way: generation → consumer	Two-way: consumers can also supply (prosumers)
Generation	Centralised large plants	Centralised + distributed (DG, renewables, storage)
Sensors / monitoring	Few sensors, manual monitoring	Sensors and PMUs throughout, self-monitoring
Fault handling	Manual restoration; failures and blackouts spread	Self-healing: automatic fault detection, isolation and restoration; adaptive protection, islanding
Control	Limited, central	Pervasive, distributed and automatic
Customer	Passive, flat tariff, monthly manual reading	Active participation: smart meters, time-of-use pricing, demand response
Equipment checking	Manual, time-based maintenance	Remote monitoring, condition-based maintenance
Efficiency and environment	High losses, hard to integrate renewables	Lower losses, easy integration of renewables and EVs

Architecture of smart grid design and functions of its components (2020 1b)



Smart grid architecture drawn for this pack: three layers

The smart grid architecture has three layers working together: (1) the **power layer** (generation, transmission, distribution, consumers); (2) the **communication layer** (WAN, NAN (neighbourhood area network), HAN (home area network) networks); (3) the **control and application layer** (control centres, data management, markets).

Component	Main function
Smart generation (central + renewables + storage)	Supplies power flexibly; renewables and storage dispatched with forecasts; generators respond to AGC (automatic generation control) signals
Smart transmission (PMU/WAMS, FACTS (flexible AC transmission systems), HVDC (high-voltage direct current), substation automation)	Wide-area monitoring and control, higher transfer capacity, stability, fast fault clearing
Smart distribution (DMS, feeder automation, DG (distributed generation), microgrids)	Self-healing feeders, Volt/VAR control, DG integration, outage management
Smart meters and AMI (advanced metering infrastructure)	Two-way metering, time-of-use pricing, remote connect/disconnect, outage detection
Customer side (HAN, smart appliances, EVs, rooftop PV)	Demand response, energy management, prosumer generation, V2G (vehicle-to-grid)
Communication network	Carries measurements and control signals securely in real time
Control centre / applications (EMS, SCADA (supervisory control and data acquisition), DMS (distribution management system), MDMS (meter data management system), analytics)	State estimation, optimal power flow, security assessment, forecasting, billing
Energy storage and power electronics	Balance variable renewables, voltage and frequency support

Key drivers of the smart grid (2022 1c)

- **Growing and changing demand** (more air-conditioning, EVs, digital loads) and need to reduce peak demand.
- **Ageing infrastructure** that must be replaced anyway; better to replace it with smart equipment.
- **Reliability and power quality**: costly blackouts (e.g. US-Canada 2003; Bangladesh 2014 and 2022 national blackouts), sensitive digital loads.
- **Integration of renewables and DG** (solar, wind) that are variable and distributed.
- **Climate change and environmental targets**: cut CO_2 by efficiency and renewables.
- **Efficiency and loss reduction**; better use of existing assets.
- **Customer empowerment and new markets**: choice, demand response, prosumers, EVs.
- **Security** (physical and cyber) and **advances in ICT** that make it affordable.

Is a smart grid necessary for Bangladesh? (2022 1c)

Yes. Reasons:

1. High system losses and electricity theft in distribution; smart/prepaid meters and AMI cut non-technical losses (prepaid smart meters are already being installed by DESCO (Dhaka Electric Supply Company), DPDC (Dhaka Power Distribution Company), BREB (Bangladesh Rural Electrification Board), WZPDCL (West Zone Power Distribution Company Ltd)).
2. Large gap between peak and off-peak demand (evening peak); DSM (demand side management) and time-of-use tariffs can flatten it and avoid costly peaking plants burning oil/diesel.
3. Frequent load shedding and grid failures (national blackouts of 2014 and 2022); WAMS (wide area monitoring system), PMUs and automation improve stability and restoration.
4. Government targets for renewables (solar home systems, rooftop net metering, solar parks, wind); a smart grid is needed to integrate variable renewables.

5. Rapid growth of demand, industrial zones and EVs (easy-bikes) that need managed charging.
6. Imported fuel (LNG, coal) is expensive; efficiency and peak reduction save foreign currency.
7. Climate commitments and vulnerability; smart grid supports clean energy and resilience to cyclones/floods (microgrids for islands and remote areas).

Challenges are cost, skills, regulation and cyber security, so it should be built step by step (prepaid meters → AMI → SCADA/DMS → WAMS → microgrids).

Main stakeholders and their functions (2022 1b)

Stakeholder	Function
Generation companies (public, IPPs, renewable developers)	Produce power, provide flexibility and ancillary services (frequency, reserve)
Transmission operator (PGCB in Bangladesh) and system operator (NLDC)	Operate the high-voltage network, balance supply and demand, ensure security and stability
Distribution utilities (DPDC, DESCO, BREB, NESCO (Northern Electricity Supply Company), WZPDCL)	Deliver power to consumers, install smart meters, DMS, integrate DG, outage management
Consumers / prosumers (residential, commercial, industrial)	Use energy, take part in demand response, generate with rooftop PV, own EVs and storage
Regulators and policy makers (BERC, Power Division, SREDA (Sustainable and Renewable Energy Development Authority))	Set tariffs, standards, rules for cost recovery, renewables and data privacy
Equipment and technology vendors, ICT/telecom providers	Supply meters, sensors, communication networks, software and cyber security
Energy service companies, aggregators, retailers	Offer energy-efficiency, demand response and EV-charging services
Researchers, standards bodies (IEEE, IEC (International Electrotechnical Commission), NIST (National Institute of Standards and Technology (USA)), universities)	Develop technologies, standards for interoperability and security, train manpower

> **Memory trick:** Smart grid = **"Eyes (sensors), Nerves (communication), Brain (control)"** added to the old grid. Comparison key words: **digital, two-way, distributed, self-healing, active customer**.

A3 Enabling technologies: information and communication; sensing, measurement, control and automation; power electronics and energy storage

Must study · Very likely · 100% · Easy

PAST-PAPER QUESTIONS

- **2023 Q1(c)** [15] Provide an overview of the following technologies required for smart grids: (i) information and communication technologies, (ii) sensing, measurement, control and automation technologies, (iii) power electronics and energy storage.
- **2022 Q2(b)** [12] Describe the roles of sensing, measurement, control and automation technologies for a smart grid.
- **2020 Q1(c)** [15] Provide an overview of the following technological requirements for a smart grid: (i) information and communication technologies, (ii) sensing, measurement, control and automation technologies, (iii) power electronics and energy storage systems.

★ REMEMBER IT

- Enabling technologies, **I-S-P**: **I**nformation & communication (networks, protocols, cyber security), **S**ensing, measurement, control & automation (smart meters, PMUs, SCADA (supervisory control and data acquisition), IEDs), **P**ower electronics & energy storage (inverters, FACTS (flexible AC transmission systems), HVDC (high-voltage direct current), batteries).

Before you start: the three technology families

To make the grid smart you need: (1) a way to **talk** (information and communication technology, ICT (information and communication technology)), (2) a way to **see and act** (sensors, meters, controllers, automation), and (3) a way to **shape and store** power (power electronics converts AC/DC and controls flows; storage keeps energy for later).

(i) Information and communication technologies (ICT)

- Provide two-way, fast, reliable and secure exchange of data between all parts of the grid; the backbone of the smart grid.
- **Wired**: optical fibre (high bandwidth, immune to interference; WAN (wide area network) backbone, substations), power line communication (PLC, uses the power cables; meters, distribution), DSL (digital subscriber line), Ethernet.
- **Wireless**: ZigBee and Bluetooth (HAN), Wi-Fi/WLAN, RF (radio frequency) mesh (NAN), WiMAX (Worldwide Interoperability for Microwave Access (IEEE 802.16)), cellular 3G/4G/5G and LTE (NAN/WAN), satellite (remote sites).
- **Networks**: HAN (home area network), NAN (neighbourhood area network)/FAN, WAN (A5); protocols and standards: IEC (International Electrotechnical Commission) 61850 (substations), DNP3 (Distributed Network Protocol 3), IEC 60870, Modbus, IEEE (Institute of Electrical and Electronics Engineers) 802.15.4, ANSI (American National Standards Institute) C12, CIM (common information model).
- **Information technology**: data management and cloud, big-data analytics, state estimation software, GIS (geographic information system), cyber security (encryption, authentication, firewalls), interoperability.

(ii) Sensing, measurement, control and automation technologies (2022 2b)

Technology	Role in the smart grid
Smart meters / AMI (advanced metering infrastructure)	Measure consumption at short intervals; enable time-of-use pricing, demand response, outage detection, remote connect/disconnect, power quality data
PMU (phasor measurement unit) **/ WAMS** (wide area monitoring system)	GPS-synchronised phasors 30-60 times/s; wide-area visibility, stability monitoring, faster state estimation (A8)
Sensors: line sensors, dynamic line rating, transformer and cable condition monitors, fault indicators, weather sensors	Detect faults, overloads, hot spots; allow higher line ratings and condition-based maintenance
IEDs (intelligent electronic devices, digital relays)	Measure, protect and control locally; communicate with SCADA
SCADA, EMS (energy management system)**, DMS** (distribution management system)	Supervise and control the whole system remotely (A5)
Substation and distribution automation	Automatic switching, reclosers, sectionalisers, FLISR (fault location, isolation and service restoration) self-healing, Volt/VAR optimisation
Advanced control methods	Adaptive protection, wide-area control, AGC (automatic generation control), intelligent agents, optimisation and AI (B8)

Together they let the grid **see** itself in real time, **decide** quickly and **act** automatically, improving reliability, efficiency and DG (distributed generation) integration.

(iii) Power electronics and energy storage

- **Power electronics**: converters/inverters that connect PV, wind, batteries, EVs and fuel cells to the grid and control their P and Q; **FACTS** devices (SVC, STATCOM (static synchronous compensator), TCSC (thyristor-controlled series capacitor), UPFC (unified power flow controller)) to control voltage and power flow; **HVDC** links for long-distance and asynchronous connections; solid-state transformers and circuit breakers; custom power devices (DVR, D-STATCOM, active filters) for power quality.
- **Energy storage**: batteries (lithium-ion, lead-acid, flow batteries, sodium-sulphur), pumped hydro, compressed air (CAES), flywheels, supercapacitors, SMES (superconducting magnetic energy storage), hydrogen/thermal storage.
- **Uses of storage**: smooth and firm up variable renewables, peak shaving and load levelling, frequency regulation and spinning reserve, voltage support, black start, backup for microgrids and islanding, deferring network upgrades.

> **Memory trick:** Three families = **"Talk, Sense-and-Act, Shape-and-Store"**.

A6 Smart meter, AMR vs AMI, how AMI works, architecture for communication between smart meter and utility server Must study

Very likely · 100% Easy

PAST-PAPER QUESTIONS

- **2023 Q3(a)** [10] What are the differences between AMR (automatic meter reading) and AMI (advanced metering infrastructure)? How do AMI meters work?
- **2022 Q3(a)** [9] What is a smart meter? Using a diagram, show the typical architecture for communication between a smart meter and the utility server.
- **2022 Q3(b)** [6] What is advanced metering infrastructure (AMI) in smart grid? Write down the differences between AMR and AMI.
- **2020 Q2(a)** [15] Using a diagram show a typical architecture for communication between a smart meter and the utility server.

★ REMEMBER IT

- **AMR** (automatic meter reading) = one-way: the meter only sends readings. **AMI** (advanced metering infrastructure) = two-way: meter ↔ utility, remote connect/disconnect, time-of-use prices, outage alerts, demand response. AMI = smart meters + communication network + meter data management system (MDMS).

Before you start: from meter reader to smart meter

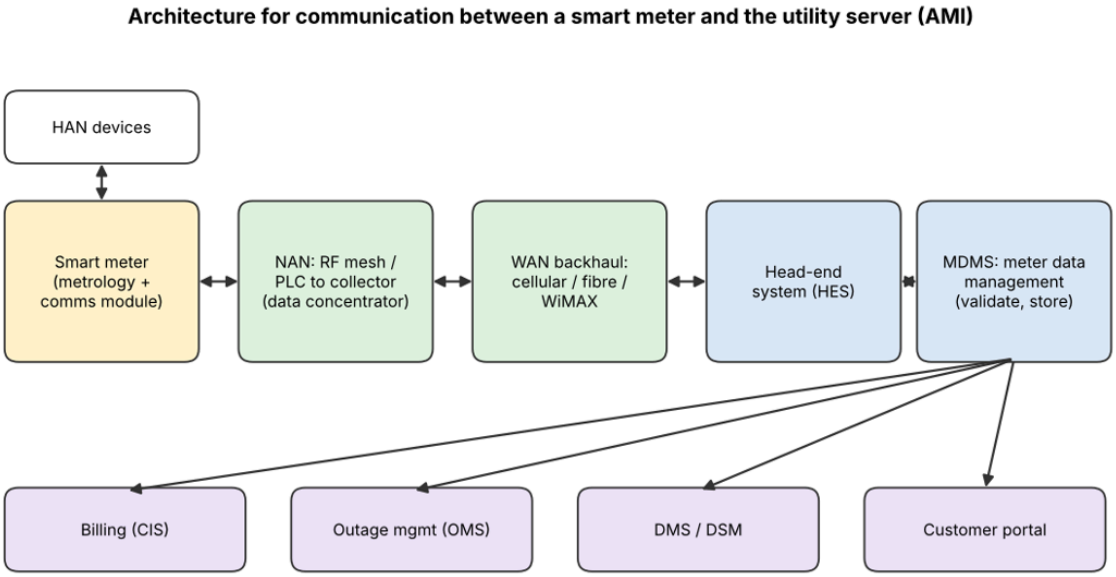
Old way: a meter reader walks to each house every month. **AMR** (automatic meter reading) lets the utility read the meter remotely, but only **one way** and mainly for billing.

AMI (advanced metering infrastructure) is a full **two-way** system: smart meters + communication network + utility software. The utility can also send commands and prices to the meter, and the meter reports outages and power quality.

Smart meter (2022 3a)

A **smart meter** is an electronic meter that measures and records electricity consumption (and voltage, demand, power factor) at short intervals (15-60 min) and communicates this information **two-way** with the utility, and often with in-home devices. Parts: metrology chip (measurement), microcontroller with memory and real-time clock, communication modules (NAN and HAN (home area network)), remote connect/disconnect relay, display, tamper sensors, power supply.

Architecture for communication between smart meter and utility server (2022 3a, 2020 2a)



Smart meter to utility server communication architecture drawn for this pack

1. **Smart meter** measures and stores interval data; talks to home devices over the HAN (ZigBee/Wi-Fi).
2. **NAN** (neighbourhood area network) **/ field area network**: meters send data by RF (radio frequency) mesh or PLC (meters can relay each other's data) to a **data concentrator (collector)** on a pole or substation.
3. **WAN** (wide area network) **backhaul**: the concentrator forwards data over cellular, fibre or WiMAX (Worldwide Interoperability for Microwave Access (IEEE 802.16)) to the utility data centre.
4. **Head-end system (HES)**: software that manages communication with all meters, collects data, sends commands (connect/disconnect, firmware update, tariff change).
5. **MDMS** (meter data management system) (meter data management system): validates, edits and estimates (VEE) and stores the meter data.
6. **Enterprise applications** use the data: billing/customer information system (CIS), outage management system (OMS), DMS (distribution management system), demand response, customer web portal and analytics.

AMR vs AMI (2023 3a, 2022 3b)

AMR (automatic meter reading)	AMI (advanced metering infrastructure)
One-way communication (meter → utility)	Two-way communication (meter ↔ utility)
Mainly automates billing: monthly or daily totals	Interval data (15-60 min), on-demand reads
Walk-by / drive-by or fixed network reading	Fixed network: meters, NAN, WAN, head-end, MDMS
No remote control	Remote connect/disconnect, load control, firmware upgrade
Flat tariffs	Time-of-use, critical-peak, real-time and prepaid pricing; demand response
No outage or power-quality information	Outage and restoration alerts, voltage and PQ monitoring, tamper detection
Customer gets only the bill	Customer gets near-real-time usage information (HAN, web)
First step (older technology)	Foundation of the smart grid

How AMI meters work (2023 3a)

1. The metrology circuit samples voltage and current thousands of times per second and computes kWh, kW, kvarh, V, pf.
2. The microcontroller time-stamps and stores interval readings in memory, applies the active tariff (time of use) and checks for events (outage, tamper, over/under voltage).
3. At scheduled times (or on request) the communication module sends the data through the NAN to the collector, then via the WAN to the head-end and MDMS.
4. The utility sends commands back the same way: price signals, demand-response events, remote disconnect/reconnect, firmware updates, prepaid credit.
5. The meter shares data with home devices through the HAN so consumers and smart appliances can react to prices.
6. Last-gasp message on power failure tells the OMS (outage management system) exactly where an outage is.

> **Memory trick: AMR = Automatic Reading (one way); AMI = Advanced, Interactive (two way)**. Path: **"Meter - Mesh - Concentrator - Backhaul - Head-end - MDMS - Billing"**.

A9 Load flow in smart grids: classical methods, old vs desired load flow, essential features, why needed and challenges Must study Very likely · 100% Easy

PAST-PAPER QUESTIONS

- **2023 Q4(c)** [10] Write down the name of the classical methods of load flow solution. What are the essential features of the load flow solution in smart grids?
- **2022 Q4(b)** [10] Make a comparison between the old and desired load flow technique in smart grids.
- **2020 Q3(a)** [10] Why are load flow studies essential in smart grid? Discuss the challenges of load flow in smart grid.

★ REMEMBER IT

- Classical load flow methods: **Gauss-Seidel** (simple, slow), **Newton-Raphson** (fast, accurate, most used), **fast decoupled** (quick approximation). Smart grids need load flow that handles two-way flow, DG (distributed generation) uncertainty, unbalanced three-phase feeders and real-time updates.

Before you start: what load flow is

Load flow (power flow) is the basic calculation of power system engineering: given the loads (P, Q) and generator outputs, find the voltage magnitude and angle at every bus, and then the power flowing in every line and the losses. It is like solving a big network of non-linear equations, so it is done by iteration.

Smart grids add DG, two-way flows, unbalanced feeders and fast-changing renewables, so the old load-flow methods need upgrading.

Why load flow studies are essential (2020 3a)

- Planning: size lines, transformers and generators for future loads and DG.
- Operation: check voltages and line flows are within limits; find losses.
- Basis for economic dispatch and optimal power flow, contingency analysis and security assessment, stability studies and state estimation.
- In smart grids: to assess DG/renewable hosting capacity, reverse flows, voltage rise, microgrid operation, EV charging impact and real-time control decisions.

Classical methods of load flow solution (2023 4c)

Method	Idea	Features
Gauss-Seidel (GS)	Update each bus voltage in turn using the latest values: $V_i = (1/Y_{ii})[(P_i - jQ_i)/V_i^* - \Sigma Y_{ij}V_j]$	Simple, little memory; slow, many iterations, poor convergence for large systems; uses Y_{bus}
Newton-Raphson (NR)	Linearise the power mismatch equations with the Jacobian and solve for corrections $[\Delta\delta; \Delta\|V\|] = J^{-1}[\Delta P; \Delta Q]$	Quadratic convergence (3-5 iterations), accurate, the industry standard; more computation per iteration
Fast decoupled (FDLF)	Uses weak coupling P-δ and Q-V, constant simplified Jacobians (B′, B″)	Very fast and memory-efficient; good for real-time and contingency studies; may fail for high R/X
DC load flow	Linear, lossless approximation: $P = B\delta$, $\|V\| = 1$	Very fast estimate of active flows; used in markets and screening

Old vs desired load flow technique for smart grids (2022 4b) and essential features (2023 4c)

Old (conventional) load flow	Desired (smart grid) load flow
Deterministic: fixed load and generation values	Probabilistic / stochastic: handles uncertain renewables, EVs and loads
Transmission level, balanced single-phase equivalent	Transmission + distribution, unbalanced three-phase, radial and meshed
Central generation, one-way flows; DG ignored	DG, storage, microgrids and two-way flows modelled (PV, PQ, droop buses)
Off-line, one snapshot	On-line, real-time, uses PMU (phasor measurement unit)/AMI data; time-series (quasi-dynamic)
Assumes low R/X; methods may diverge on distribution feeders	Robust for high R/X feeders (backward-forward sweep, current injection)
Single objective: find state	Integrated with optimisation (OPF, Volt/VAR), contingency and stability analysis, adaptive/intelligent
Moderate size	Very large scale; parallel/distributed computing, fast

Essential features of load flow in smart grids: fast and robust convergence; handles unbalanced three-phase distribution and high R/X; models DG, storage, power electronics and microgrid controls; stochastic/probabilistic to deal with intermittency; works in real time with PMU and smart meter data; distributed computation; integrates with optimisation and security tools.

Challenges of load flow in smart grid (2020 3a)

1. Intermittent, uncertain renewables and EV loads (need probabilistic and time-series studies).
2. Two-way power flow and voltage rise with DG.
3. Unbalanced, radial distribution networks with high R/X (NR and FDLF (fast decoupled load flow) may diverge).
4. Very large size: transmission + distribution + microgrids → heavy computation.
5. Modelling of inverters, storage and their controls (droop, Q-limits).
6. Real-time requirements with huge data from AMI (advanced metering infrastructure) and PMUs; data quality and communication delays.

7. Islanding/microgrid operation without a single slack bus.

> **Memory trick:** Classical = **"GS slow, NR strong, Fast Decoupled fast, DC rough"**. Smart load flow = **"Stochastic, Three-phase, DG-aware, Real-time, Robust"**.

A4 Distributed generation (DG) and microgrid: definition, advantages, configuration, grid-connected and islanded operation Must study

Very likely · 83% Easy

PAST-PAPER QUESTIONS

- **2023 Q2(a)** [10] Define the term "distributed generation (DG)". What are the advantages of DG (distributed generation)?
- **2023 Q2(b)** [10] Draw the typical configuration of a micro-grid. Describe the grid-connected and islanded operation of the micro-grid.
- **2022 Q2(a)** [11] Define the term "distributed generation (DG)". What are the advantages of DG?
- **2022 Q3(d)** [8] Draw the configuration of a typical microgrid and describe its operation.

LEARN IT FROM ZERO

Distributed generation and microgrids, from zero

Distributed generation (DG) = small generators (kW to a few MW) placed close to the loads, in the distribution network: rooftop PV, small wind, CHP (combined heat and power), fuel cells, diesel sets. Benefits: lower line losses, less transmission needed, better reliability, uses local renewables, faster to build.

A **microgrid** is a small piece of the grid (a campus, village, factory) with its own DG, storage and loads, and a **point of common coupling (PCC)** switch to the main grid. **Grid-connected mode**: the switch is closed; the main grid fixes voltage and frequency and absorbs any surplus or deficit. **Islanded mode**: on a grid fault the switch opens; now the microgrid's own controllers (and storage) must hold voltage and frequency and shed low-priority loads if needed. When the grid is back, it resynchronises and recloses.

Before you start: small power near the user

Instead of only a few giant power stations far away, many **small generators** are placed close to where power is used: rooftop solar panels, small wind turbines, biogas or diesel gensets, fuel cells, micro-turbines. That is **distributed generation (DG)**.

Group some DGs, batteries and loads together with their own controller and a switch to the main grid, and you get a **microgrid**: a small grid that can work with the big grid or alone (islanded) when the big grid fails.

Distributed generation: definition (2023 2a, 2022 2a)

Distributed generation (DG) is electric power generation within distribution networks or on the customer side of the network, close to the point of use, typically from a few kW up to about 50 MW, not centrally planned or dispatched. Technologies: solar PV, wind turbines, small hydro, biomass/biogas, fuel cells, micro-turbines, reciprocating (diesel/gas) engines, combined heat and power (CHP), and storage.

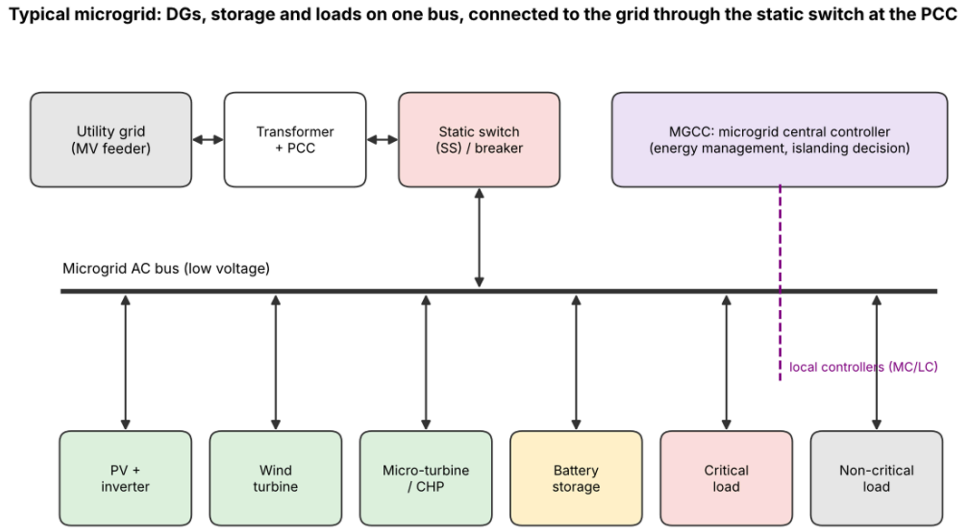
Advantages of DG

1. **Reduced transmission and distribution losses** because power is produced near the load.

2. **Improved voltage profile** along feeders and better power quality.
3. **Higher reliability**: backup supply and islanding (microgrids) during grid outages.
4. **Deferral of investment** in new transmission/distribution lines and large plants; relieves congestion.
5. **Uses renewable energy** → lower emissions and fuel import; supports climate goals.
6. **High overall efficiency with CHP** (waste heat used, up to 80-90 %).
7. **Short installation time, modular, low capital risk**; easy to site.
8. **Electrification of remote areas** and islands (solar home systems, mini-grids in Bangladesh).
9. Peak shaving and ancillary services; customer choice and energy independence.

> **Side note:** Challenges of DG: reverse power flow, voltage rise, changes in fault level and protection, islanding safety, harmonics from inverters, intermittency - handled by active distribution management and microgrids.

Microgrid configuration and operation (2023 2b, 2022 3d)



Typical microgrid configuration drawn for this pack

A **microgrid** is a group of interconnected loads and distributed energy resources (DGs and storage) within clearly defined electrical boundaries that acts as a single controllable entity with respect to the grid and can operate in both **grid-connected** and **island** modes.

Components: DG units (PV, wind, micro-turbine, fuel cell, diesel) with power-electronic interfaces; energy storage; critical and non-critical loads; the **point of common coupling (PCC)** with a **static switch** (fast thyristor/IGBT switch or breaker) to the utility; **microsource controllers** (local controllers at each DG) and the **microgrid central controller (MGCC)** for energy management.

Grid-connected mode: the static switch is closed. The utility grid fixes voltage and frequency; DGs run in constant-power (PQ) mode at their economic/maximum output; surplus is exported and deficits imported; storage charges or shaves peaks; the microgrid can sell power and provide services (demand response, reactive power).

Islanded (autonomous) mode: when a fault or poor power quality is detected on the utility side, the static switch opens within a few cycles and the microgrid separates. Now it must create its own voltage and frequency: one or more DGs/storage inverters switch to **voltage-frequency (V/f) control or droop control** (frequency falls slightly as load rises, so units share load); storage covers

the sudden mismatch; non-critical loads are shed if generation is insufficient so that critical loads (hospitals, control rooms) stay supplied. When the grid recovers, the MGCC (microgrid central controller) **resynchronises** (matches voltage, frequency, phase) and recloses the static switch.

> **Memory trick:** DG advantages = **"Less Loss, Better Voltage, More Reliability, Defer Investment, Green"**. Microgrid: **PCC (point of common coupling) + static switch + DGs + storage + loads + MGCC; PQ mode on-grid, V/f droop off-grid**.

A7 Demand side management (DSM): load shifting, valley filling, peak clipping (load-shape curves) Must study Very likely · 83% Easy

PAST-PAPER QUESTIONS

- **2023 Q3(b)** [15] What is meant by demand side management? Describe how the following techniques modify the load consumption patterns: (i) load shifting, (ii) valley filling, (iii) peak clipping.
- **2022 Q3(c)** [12] What is meant by demand side management? Describe how the following techniques modify the load consumption patterns: (i) load shifting, (ii) valley filling, (iii) peak clipping.

LEARN IT FROM ZERO

Demand side management, from zero

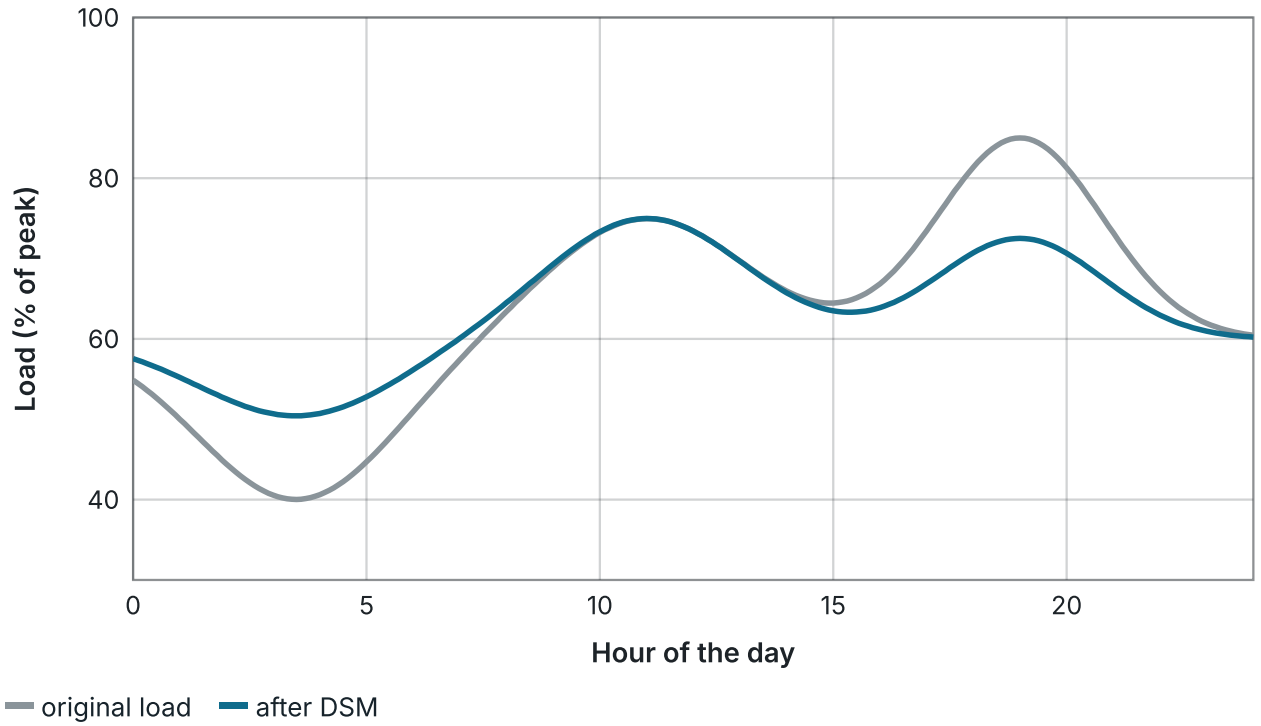
Building a power plant just to cover the evening peak is very expensive. Instead, **demand side management (DSM)** changes *when* and *how much* customers use, so the daily load curve becomes flatter. The classic load-shape objectives:

- **Peak clipping**: cut load at the peak (utility switches off water heaters/ACs for a while). Peak goes down, energy goes down a little.
- **Valley filling**: add load at night when demand is low (EV charging, ice storage). Valley comes up, energy goes up.
- **Load shifting**: move load from the peak to the valley (run washing machines at night with time-of-use tariffs). Same energy, flatter curve.
- Others: strategic conservation (efficiency, less energy everywhere), strategic load growth, flexible load shape.

Move the sliders: the shaded curve is the original, the coloured one is after DSM (demand side management).

INTERACTIVE **DSM: how each technique changes the daily load curve**

Move the sliders and watch the curve change.



Load shifting moves the same energy from the evening peak to the night valley, so the curve gets flatter and the load factor rises.

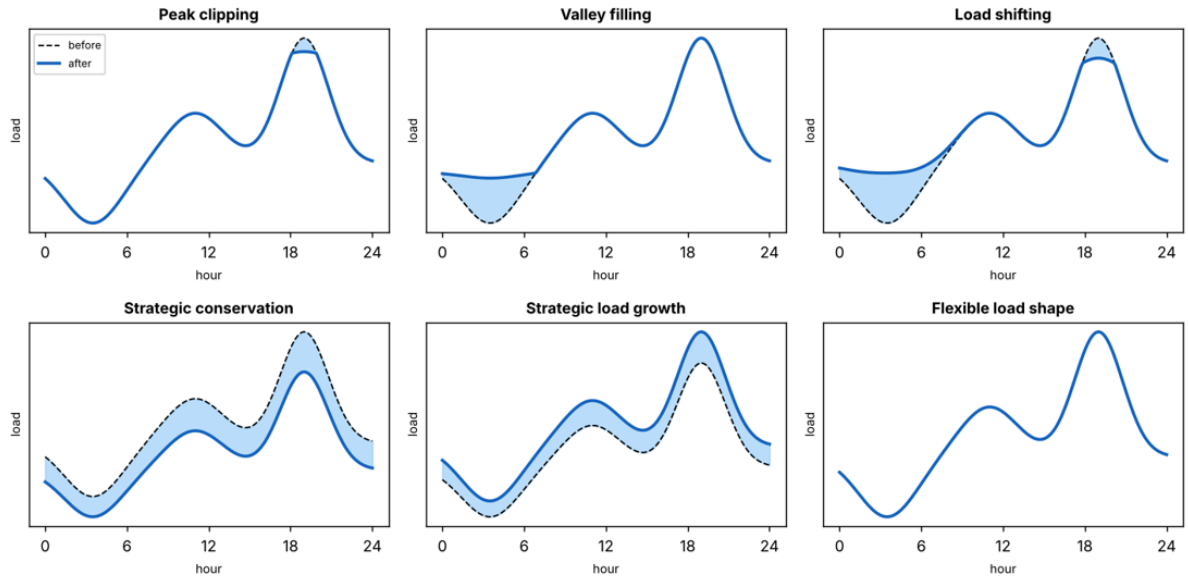
Before you start: shaping the load curve

The load curve (power demand vs hour of the day) has a tall evening peak and a deep night valley. The utility must build enough plants for the peak, which then sit idle most of the day - very expensive.

DSM (demand side management) changes **when** and **how much** customers use, so the curve becomes flatter. Think of six shapes: cut the top, fill the bottom, move load from top to bottom, use less overall, add useful load, or keep load flexible.

Demand side management (2023 3b, 2022 3c)

Demand side management (DSM) is the planning, implementation and monitoring of utility activities designed to influence customers' use of electricity in ways that produce desired changes in the utility's load shape - the time pattern and magnitude of demand. Tools: tariffs (time-of-use, critical-peak, real-time pricing), direct load control, interruptible contracts, energy-efficiency programmes, storage and smart appliances, all made practical by smart meters and AMI (advanced metering infrastructure). Benefits: lower peak, less need for peaking plants and network upgrades, lower cost and emissions, better reliability, and lower bills for customers.



The six DSM load-shape objectives (dashed = before, blue = after) drawn for this pack

Technique	How it modifies the load curve	Means	Effect
(i) Load shifting	Moves load from peak hours to off-peak hours; total energy about the same	Time-of-use tariffs; thermal (ice/water) storage; scheduling washing machines, pumps, EV charging, industrial processes at night; battery storage	Lower peak AND filled valley; flatter curve; best use of base-load plants
(ii) Valley filling	Builds load in off-peak periods (night) only	Cheap night tariffs; EV charging at night; off-peak water heating and pumping; charging storage	Raises the valley and the load factor; lowers average cost of supply since spare base capacity is used
(iii) Peak clipping	Reduces load only during peak hours	Direct load control (utility switches off air-conditioners, water heaters, pumps for short periods); interruptible tariffs; demand response events; critical-peak pricing	Cuts the peak demand, avoids building/running costly peaking plants; total energy reduced slightly
Strategic conservation	Reduces load at all hours	Efficient appliances, LED lighting, insulation	Whole curve lower
Strategic load growth	Increases load in a planned way	Electrification (EVs, heat pumps, induction cooking)	Whole curve higher but well-utilised
Flexible load shape	Load can be varied when needed	Contracts with flexible customers, smart appliances	Reliability tool

Memory trick: Clip the top, Fill the bottom, Shift top to bottom.

A2 Active vs passive distribution network; active distribution management scheme (block diagram) Must study Very likely · 83% Easy

PAST-PAPER QUESTIONS

- **2023 Q1(b)** [10] What are the differences between an active and a passive distribution network? Draw the block diagram of an active distribution system and describe its operation.
- **2022 Q1(a)** [15] What are the differences between an active and a passive distribution network? Describe the operation of an active distribution management scheme with necessary block diagrams.

UNDERSTAND **Read this to understand; you do not need to write it**

A **passive** distribution network only delivers power one way from the substation to loads; nothing is controlled after the feeder breaker. Once DG (rooftop solar, small generators) is connected, power can flow backwards and voltages can rise, so the network must become **active**: it measures, communicates and controls generators, loads, tap changers and storage in real time from a distribution management system (DMS).

Before you start: passive vs active

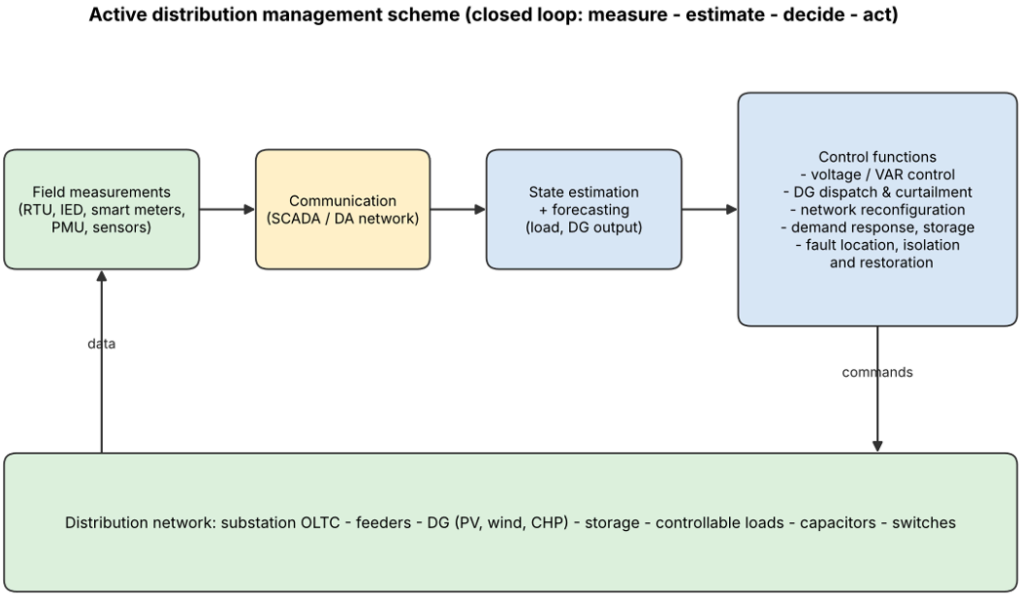
A **passive** distribution network is like a one-way street: power comes from the substation and flows down radial feeders to loads. Designed once with the worst case and then left alone ("fit and forget").

An **active** network has generators (rooftop PV, wind, CHP (combined heat and power)), storage and controllable loads on the feeders, so power can flow back up. It must therefore be **watched and steered** continuously, like traffic control on a two-way road.

Active vs passive distribution network (2023 1b, 2022 1a)

Passive distribution network	Active distribution network
No (or negligible) generation; only loads	Has DG (distributed generation), storage, EVs, controllable loads
Unidirectional power flow (substation → loads)	Bidirectional power flow
Voltage falls along the feeder; fixed tap and capacitor settings	Voltage can rise near DG; needs coordinated, dynamic voltage control
"Fit and forget" planning for worst case	Real-time monitoring, control and optimisation
Simple overcurrent protection (radial)	Adaptive protection (fault current from DG, two-way flows), anti-islanding
Little communication and measurement	Extensive sensors, communication, SCADA (supervisory control and data acquisition)/DMS
Consumers passive	Consumers active (prosumers, demand response)
Low DG hosting capacity	Higher DG hosting capacity, better asset use

Block diagram and operation of an active distribution management scheme



Active distribution management scheme drawn for this pack

Operation (a closed control loop that repeats every few seconds to minutes):

1. **Measure:** RTUs, IEDs, smart meters, PMUs and sensors measure voltages, currents, power flows, switch status, DG output at substations and along feeders.
2. **Communicate:** data go through the SCADA / distribution automation network to the distribution management system (DMS) in the control centre.
3. **Estimate and forecast:** distribution state estimation fills gaps in the measurements and gives the complete network state; load and DG (solar/wind) output are forecast.
4. **Decide (control functions):** coordinated voltage/VAR control (on-load tap changer, capacitors, DG reactive power); DG dispatch or curtailment and storage scheduling; network reconfiguration to reduce losses or relieve overloads; demand response; fault location, isolation and service restoration (FLISR, self-healing); protection setting adaptation.
5. **Act:** commands are sent back to tap changers, switches, DG inverters, storage and controllable loads.
6. The new state is measured again, closing the loop. Operators supervise through the HMI (human-machine interface).

Side note: The control can be **centralised** (one DMS (distribution management system) decides), **decentralised** (local controllers at DGs and substations) or **hierarchical** (local + central), and in research also by multi-agent systems (A11).

Memory trick: Active network loop: **"Measure - Send - Estimate - Decide - Act"**. Passive = **"fit and forget"**.

A5 Smart grid communication: SCADA, HAN/NAN/WAN, and where Bluetooth, ZigBee, WLAN, WPAN and WiMAX are used Must study

Very likely · 100% Medium

PAST-PAPER QUESTIONS

- **2023 Q2(c)** [15] What is meant by SCADA (supervisory control and data acquisition)? What are the advantages of a SCADA integrated system? Show the possible communication infrastructure for the smart grid including HAN (home area network), NAN (neighbourhood area network) and WAN (wide area network).
- **2022 Q2(c)** [12] Define HAN, NAN and WAN. Compare the conventional and smart meter. Write down the application areas of the following communication technologies in smart grid: Bluetooth, WiMAX (Worldwide Interoperability for Microwave Access (IEEE 802.16)) and WLAN (wireless local area network (Wi-Fi)).
- **2020 Q2(b)** [15] Write down the application areas of the following communication technologies in smart grid: Bluetooth, ZigBee, WLAN, WPAN (wireless personal area network), WiMAX.

★ REMEMBER IT

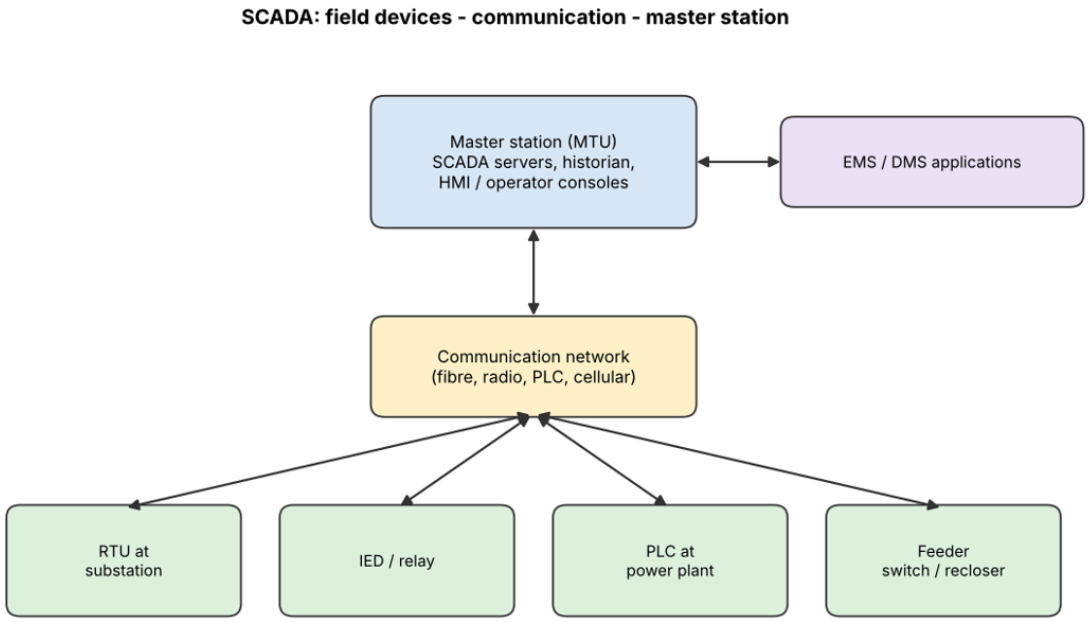
- Network sizes: **HAN** (home area: ZigBee, Wi-Fi, Bluetooth; metres) → **NAN** (neighbourhood: meters to collector; RF (radio frequency) mesh, PLC, WiMAX; hundreds of m to km) → **WAN** (utility backbone: fibre, cellular, WiMAX; tens of km).
- Where each fits: Bluetooth = very short (meter to phone); ZigBee = low power home devices and meters; WLAN = home/substation LAN; WPAN = personal devices; WiMAX = long range NAN/WAN backhaul.
- SCADA = Supervisory Control And Data Acquisition: RTUs in substations send data to the control centre; operators supervise and switch remotely.

Before you start: how the grid talks

SCADA (supervisory control and data acquisition) is the classic remote-control system of a utility: little computers in the field (RTUs) read meters and switch positions, send them over a network to a master station, and the operator can open/close breakers from the control room.

Smart grid communication is organised by distance, like phone networks: **HAN** inside one home, **NAN** across a neighbourhood, **WAN** across the country to the utility.

SCADA and its advantages (2023 2c)



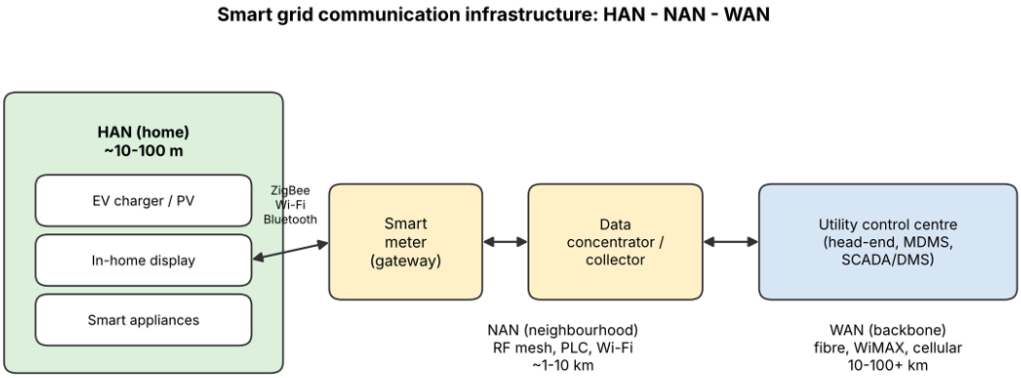
SCADA structure drawn for this pack

SCADA is a computer-based system that gathers real-time data from remote field devices, presents it to operators and lets them control equipment remotely. Components: **field devices** (sensors, CTs/PTs, breakers), **RTUs** (remote terminal units) / **IEDs** / PLCs that acquire data and execute commands, a **communication network** (fibre, radio, PLC, cellular), and the **master station (MTU)** with servers, historian database and **HMI** (human-machine interface) screens. Functions: data acquisition, monitoring and alarms, supervisory control, data logging, trending; EMS (energy management system)/DMS applications run on top.

Advantages of a SCADA-integrated system:

1. Real-time monitoring of the whole network from one place; faster fault detection and alarms.
2. Remote control of breakers, tap changers, capacitors → fewer site visits and staff.
3. Faster restoration after faults, less outage time (higher reliability indices).
4. Better efficiency: economic dispatch, loss reduction, voltage control via EMS/DMS.
5. Historical data for planning, maintenance and analysis; condition monitoring.
6. Improved safety of personnel and equipment.
7. Integration with smart grid applications (AMI, DMS (distribution management system), OMS (outage management system), WAMS (wide area monitoring system)).

HAN, NAN and WAN (2023 2c, 2022 2c)



Possible communication infrastructure for the smart grid drawn for this pack

Network	Meaning and coverage	Connects	Technologies
HAN (home area network)	Inside a home/building; ~10-100 m; low data rate	Smart meter, smart appliances, thermostats, in-home display, EV charger, rooftop PV	ZigBee, Bluetooth, Wi-Fi, HomePlug (PLC), Z-Wave
NAN (neighbourhood area network; also FAN (field area network), field area network)	A neighbourhood or feeder; ~1-10 km	Many smart meters to a data concentrator/collector; distribution automation devices	RF mesh, Wi-Fi mesh, PLC, WiMAX, cellular
WAN (wide area network)	Utility-wide backbone; 10-100+ km; high data rate	Collectors, substations and power plants to control centres	Optical fibre, WiMAX, cellular LTE (Long Term Evolution (4G cellular))/5G, microwave, satellite

Conventional meter vs smart meter (2022 2c)

Conventional meter	Smart meter
Electromechanical (induction disc) or simple electronic	Fully electronic with microprocessor and memory
Records only total kWh	Records kWh, kW demand, voltage, power factor, at 15-30 min intervals
Manual reading once a month	Automatic remote reading; two-way communication
Flat tariff only	Time-of-use, real-time, prepaid tariffs
No outage or tamper information	Outage and tamper alarms, remote connect/disconnect
Consumer sees nothing until the bill	Consumer can see usage in real time (in-home display, app)

Application areas of communication technologies (2022 2c, 2020 2b)

Technology	What it is	Smart grid applications
Bluetooth (IEEE 802.15.1)	Short-range (~10 m) low-power wireless, 2.4 GHz, ~1-3 Mbit/s	HAN: linking in-home displays, phones and appliances to the smart meter; configuring meters and relays on site; wearable/handheld tools for field crews
ZigBee (IEEE 802.15.4)	Low-power, low-cost, low-data-rate (250 kbit/s) mesh network, 10-100 m	HAN: smart meter to appliances, thermostats, load-control switches (Smart Energy Profile); demand response; street-light control; sensor networks
WLAN (Wi-Fi, IEEE (Institute of Electrical and Electronics Engineers) 802.11)	Wireless LAN, 54 Mbit/s to Gbit/s, ~100 m	HAN/BAN: home and building energy management; substation automation LAN; NAN mesh for meter collection; EV charging stations; field crew access
WPAN (wireless personal area network, IEEE 802.15 family incl. Bluetooth, ZigBee, 6LoWPAN)	Very short-range networks around a person/device	Connecting meters, sensors and appliances in HAN; monitoring sensors on equipment; personal devices of field staff
WiMAX (IEEE 802.16)	Broadband wireless metropolitan network, up to ~50 km, tens of Mbit/s	NAN/WAN backhaul from data concentrators to the utility; distribution automation and SCADA links to remote substations; wide-area monitoring; video surveillance

Memory trick: Distance ladder: **HAN home < NAN neighbourhood < WAN wide**. Technologies: **"Blue and Zig at home, Wi-Fi in the building, WiMAX across the town"**.

A10 Performance analysis tools: contingency analysis, security-constrained OPF, DSOPF, static security assessment (SSA), congestion management; GIS

Must study | Very likely · 100% | Medium

PAST-PAPER QUESTIONS

- **2023 Q4(a)** [10] Why is contingency analysis important for power system studies? Describe the term "security-constrained optimal power flow" with an example.
- **2023 Q4(b)** [5] Write down the necessity of GIS (geographic information system) and Google mapping tools in smart grids.
- **2022 Q4(c)** [5] Write a short note on static security assessment (SSA) of smart power systems.
- **2020 Q3(b)** [20] Provide an overview of the following performance analysis tools for smart grid design: (i) DSOPF (dynamic stochastic optimal power flow), (ii) SSA (static security assessment), (iii) congestion, (iv) contingency.

★ REMEMBER IT

- Contingency analysis = "what if line X or generator Y trips?" (N−1). SCOPF (security-constrained optimal power flow) = optimal power flow that also stays safe for every listed contingency. DSOPF = dynamic stochastic OPF (handles uncertainty over time). SSA = static security assessment: checks voltage and line limits after each contingency. Congestion = a line is at its limit, so cheaper power cannot reach the load.

Before you start: checking the grid is safe and cheap

A grid must survive the sudden loss of any one line or generator (the **N-1** rule). **Contingency analysis** asks "what if this element trips?" for every element. **Security assessment** uses these answers to say whether the present state is safe.

Optimal power flow (OPF) finds the cheapest generator settings that respect all limits. Make it respect the limits **after** contingencies too and it is **security-constrained OPF (optimal power flow)**.

N-1 means: with N elements, the system must still be OK with any one (1) of them out.

Contingency analysis and why it is important (2023 4a, 2020 3b)

Contingency analysis is the study of the effect of the outage of a component (line, transformer, generator) on the power system: for each credible contingency it runs a (fast) load flow and checks for line overloads, voltage violations and islanding. Contingencies are screened and ranked by severity indices so that only critical ones are studied in detail.

- Ensures the system remains secure (N-1, N-2) and prevents cascading outages and blackouts.
- Tells operators which outages are dangerous and what preventive or corrective actions (re-dispatch, switching, load shedding) are needed.
- Basis for security assessment, SCOPF, transmission planning and maintenance scheduling.
- In smart grids, needed more often (in real time) because renewables and two-way flows change conditions quickly.

Security-constrained optimal power flow (SCOPF) with example (2023 4a)

OPF minimises an objective (generation cost or losses) subject to the power flow equations and limits (generator P, Q limits, bus voltage limits, line thermal limits). **SCOPF** adds the constraints of each credible contingency: the chosen operating point must keep all flows and voltages within (emergency) limits not only in the base case but also after any single contingency (preventive SCOPF), or must be correctable by allowed post-contingency actions (corrective SCOPF).

$\min \Sigma C_i(P_{Gi})$ subject to: $g_0(x_0, u) = 0$, $h_0(x_0, u) \leq 0$ (base case); $g_k(x_k, u) = 0$, $h_k(x_k, u) \leq 0$ for every contingency k

Example: two generators feed a load centre of 300 MW through two parallel lines of 200 MW rating each. The cheap generator G1 (Tk 3/kWh) is at the far end, the costly G2 (Tk 8/kWh) at the load centre. Plain OPF sends all 300 MW from G1 through the two lines (150 MW each). If one line trips, the other must carry 300 MW > 200 MW → overload and cascading outage. SCOPF limits G1 so that the remaining line carries ≤ 200 MW after the trip: G1 = 200 MW and G2 = 100 MW. The cost is higher, but the system stays secure under N-1.

Static security assessment (SSA) (2022 4c, 2020 3b)

SSA evaluates whether the power system, in its steady state after a contingency, will operate within all limits (line flows, bus voltages, generator limits) - ignoring the transient swings in between. Steps: (1) obtain the present state from state estimation; (2) select a list of credible contingencies; (3) run contingency (load flow) analysis for each; (4) check limit violations and compute severity indices; (5) classify the system as **secure** (normal), **alert/insecure** (a contingency would cause violation) or **emergency**; (6) suggest preventive/corrective control. In smart grids, SSA must be fast (using PMU (phasor measurement unit) data, DC/decoupled load flow, AI/pattern recognition) because conditions change rapidly.

DSOPF (2020 3b)

DSOPF (dynamic stochastic optimal power flow), proposed by Momoh for smart grids, extends OPF to handle **time** (dynamic: the decision now affects future states) and **uncertainty** (stochastic: random loads, renewable output, outages). It combines OPF with adaptive dynamic programming / approximate dynamic programming so the optimiser learns and adapts in real time. Uses: optimal dispatch with renewables and storage, adaptive Volt/VAR control, reconfiguration, demand response, self-healing and resilient operation. Advantages: predictive, adaptive, handles variable resources; disadvantage: heavy computation.

Congestion (2020 3b)

Congestion occurs when a transmission line or transformer reaches its thermal, voltage or stability limit, so the desired (cheapest) power transfers cannot all be made. **Congestion management**: re-dispatch of generation (out-of-merit), FACTS (flexible AC transmission systems) and phase-shifting transformers to re-route flows, demand response and storage at the receiving end, locational marginal pricing and congestion charges/financial transmission rights in markets, network reinforcement, dynamic line rating. Smart grid tools (PMUs, DLR (dynamic line rating) sensors, DSM (demand side management)) relieve congestion at lower cost.

Necessity of GIS and Google mapping tools in smart grids (2023 4b)

- Show every asset (poles, lines, transformers, meters, DGs) on a map with its attributes: an accurate network model for load flow and DMS (distribution management system).
- Outage management: locate faults and affected customers on the map; dispatch the nearest crew with the best route.
- Planning: route new lines, site DG (distributed generation)/solar and substations considering land, terrain, flood zones.
- Vegetation management, asset maintenance and theft detection.
- Visualising real-time data (loads, outages, EV chargers) for operators and customers; emergency/disaster response.

> **Memory trick: Contingency = "what if one trips?"; SSA = "are we safe after it?"; SCOPF = cheapest that is still safe; DSOPF = OPF + time + uncertainty; congestion = traffic jam on lines**.

A8 Phasor measurement unit (PMU) and synchronised phasor measuring system (WAMS) Must study Very likely · 83% Medium

PAST-PAPER QUESTIONS

- **2023 Q3(c)** [10] What are the key features of a phasor measurement unit (PMU)? Explain the operation of a PMU (phasor measurement unit) with necessary block diagrams.
- **2022 Q4(a)** [10] What is a phasor measurement unit in smart grid? Discuss a synchronized phasor measuring system with conceptual diagram.

UNDERSTAND **Read this to understand; you do not need to write it**

A **PMU** (phasor measurement unit) measures voltage and current phasors (magnitude **and** angle) 30–60 times a second, each stamped with GPS (Global Positioning System) time to within a microsecond. Because all PMUs share the same clock, a control centre can line up angles from stations hundreds of km apart and see the whole grid's state at once (a **synchrophasor** or wide-area measurement system, WAMS (wide area monitoring system)).

Block diagram to draw: CT/PT input → anti-aliasing filter → A/D converter (sampling clock locked to GPS via phase-locked oscillator) → phasor microprocessor (DFT) → time-stamped output → communication → PDC (phasor data concentrator) → control centre.

Before you start: measuring the grid's heartbeat in sync

AC voltage is a sine wave; a **phasor** describes it by its size (magnitude) and its position in time (phase angle). Comparing phase angles at two far-apart substations tells you how stressed the grid between them is (bigger angle difference = more power flowing, closer to instability).

But to compare angles, both measurements must be taken at **exactly the same moment**. A **PMU** uses GPS satellite time (accurate to 1 microsecond) to time-stamp every measurement, so measurements from the whole country can be lined up. Old SCADA (supervisory control and data acquisition) gives one slow, unsynchronised reading every 2-4 seconds; a PMU gives 30-60 synchronised readings per second.

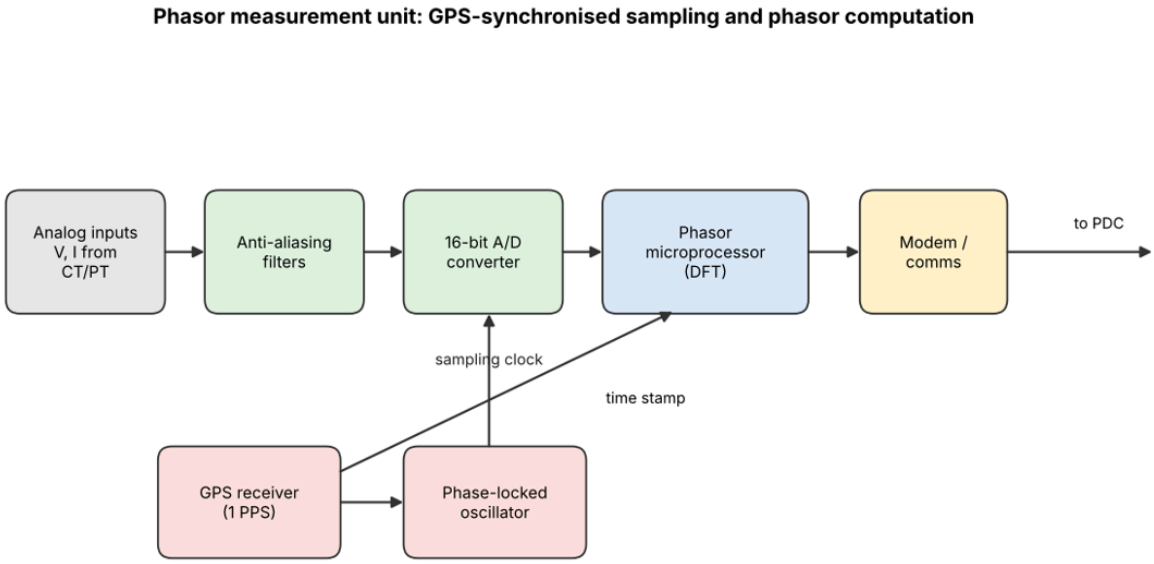
PMU: definition and key features (2023 3c, 2022 4a)

A **phasor measurement unit (PMU)** is a device that measures the voltage and current phasors (magnitude and phase angle), frequency and rate of change of frequency (ROCOF) at a bus, synchronised to a common time source (GPS), and streams them as time-stamped **synchrophasors** (IEEE C37.118 standard).

- **GPS time synchronisation** (1 µs accuracy; 1 µs = 0.018° at 50 Hz), so angles from different locations are directly comparable.
- **High reporting rate**: 25/50 frames per second at 50 Hz (30/60 at 60 Hz) vs 2-4 s for SCADA - captures dynamics and oscillations.
- Measures positive-sequence voltage and current phasors, frequency, ROCOF (rate of change of frequency), also individual phases.
- Direct measurement of the phase angle (SCADA cannot measure angles; state estimation had to calculate them).

- High accuracy (total vector error < 1 %), standard data format, can be stand-alone or built into relays/recorders.
- Enables wide-area monitoring, protection and control (WAMPAC).

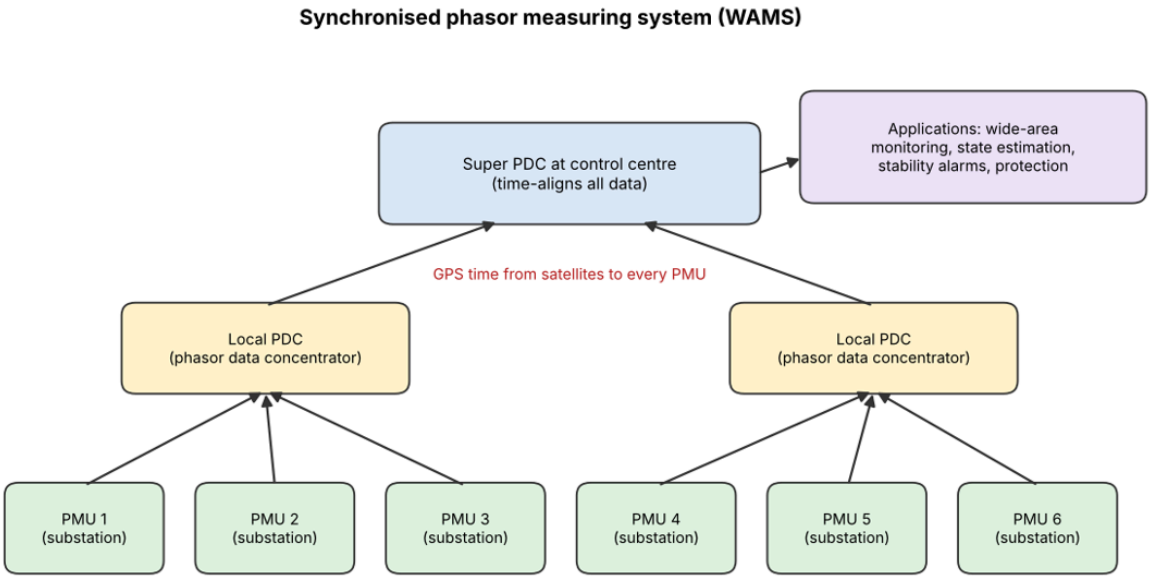
Operation of a PMU (block diagram)



PMU block diagram drawn for this pack

1. **Analog inputs**: three-phase voltages and currents from potential and current transformers (PT/CT).
2. **Anti-aliasing filters** remove frequencies above half the sampling rate so the digital samples are correct.
3. **A/D converter** (16-bit) samples the waveforms; the **sampling clock** comes from a **phase-locked oscillator** locked to the **GPS receiver's one-pulse-per-second (1 PPS (pulse per second))** signal, so every PMU samples at the same instants.
4. **Phasor microprocessor** computes the phasor of each signal with a recursive DFT (discrete Fourier transform) over one cycle, then the positive-sequence phasor, frequency and ROCOF.
5. Each result is **time-stamped** with the GPS time (UTC).
6. The **modem/communication** module sends the data frames to a PDC (phasor data concentrator).

Synchronised phasor measuring system / WAMS (2022 4a)



Synchronised phasor measuring system (WAMS) drawn for this pack

PMUs at many substations send data to **local PDCs (phasor data concentrators)**, which align measurements with the same time stamp, check quality and forward them to a **super PDC** at the control centre. Applications use the coherent snapshot of the whole grid: wide-area visualisation (angle differences across the network), real-time monitoring of oscillations and voltage stability, improved state estimation (direct linear estimation), post-disturbance analysis, model validation, adaptive and wide-area protection, islanding detection, and control of FACTS (flexible AC transmission systems)/HVDC.

> **Memory trick:** PMU = **"GPS + Filter + A/D + DFT (discrete Fourier transform) + Time stamp + Send"**. WAMS = **PMUs → PDC → super PDC → applications**.

A11 Multi-agent systems (MAS): IDAPS and decentralised meter reading

Must study | Very likely · 83% | Medium

PAST-PAPER QUESTIONS

- **2023 Q4(d)** [10] Design a decentralized meter reading system using the multi-agent approach.
- **2022 Q4(d)** [10] What is meant by a multi-agent system (MAS)? Develop an intelligent distributed autonomous power system (IDAPS) using the MAS (multi-agent system) approach.

UNDERSTAND **Read this to understand; you do not need to write it**

A multi-agent system is a team of small software programs (agents), each with its own job and a bit of intelligence, that talk to each other to reach a goal without one central boss. In IDAPS (intelligent distributed autonomous power system) you would have a control agent (watches the grid, detects outages), DER (distributed energy resource) agent (controls each generator/storage), user agent (each customer's loads and preferences) and database agent (stores data). On a grid fault they island the area and keep critical loads supplied by themselves.

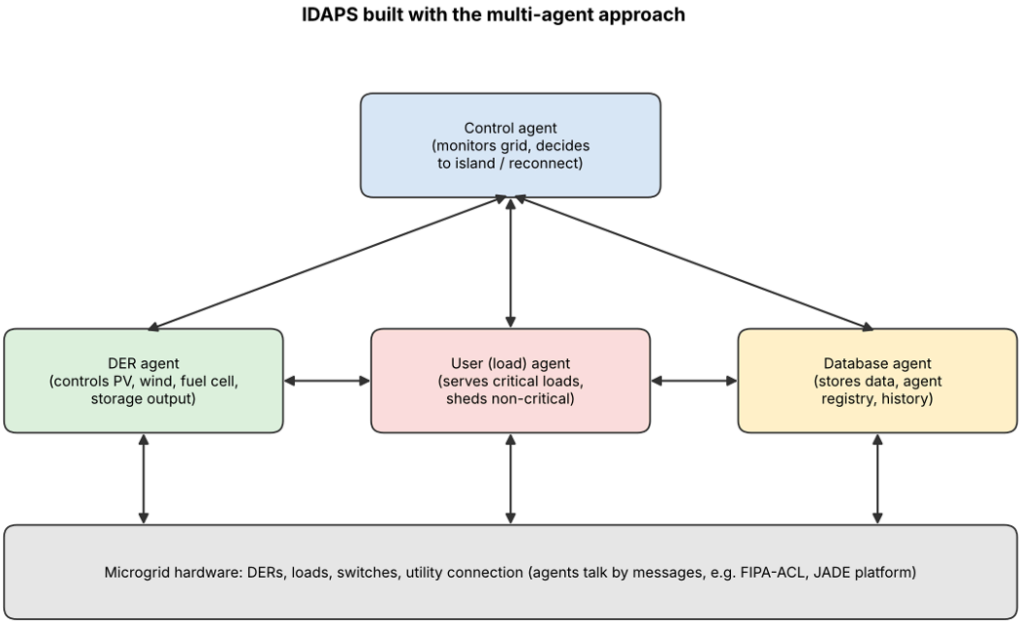
Before you start: many small brains instead of one big one

In a **multi-agent system (MAS)** each part of the grid (a DG (distributed generation), a load, a meter) gets its own small software "agent" that can see its local situation, decide and act by itself, and chat with other agents. Together they solve the problem without one central computer. If one agent fails, the others continue: no single point of failure.

Multi-agent system (2022 4d)

An **agent** is a software (or hardware) entity situated in an environment that is **autonomous** (acts without direct human control), **reactive** (senses and responds to changes), **pro-active** (pursues its goals) and **social** (communicates and cooperates with other agents). A **multi-agent system (MAS)** is a group of such agents that interact (through messages in an agent communication language such as FIPA-ACL, on platforms like JADE (Java Agent DEvelopment framework)) to achieve goals that are beyond any single agent. Advantages for smart grids: distributed and scalable, flexible (plug-and-play DG), robust to failures, fast local decisions, suits distributed resources and microgrids.

IDAPS (intelligent distributed autonomous power system) using the MAS approach (2022 4d)



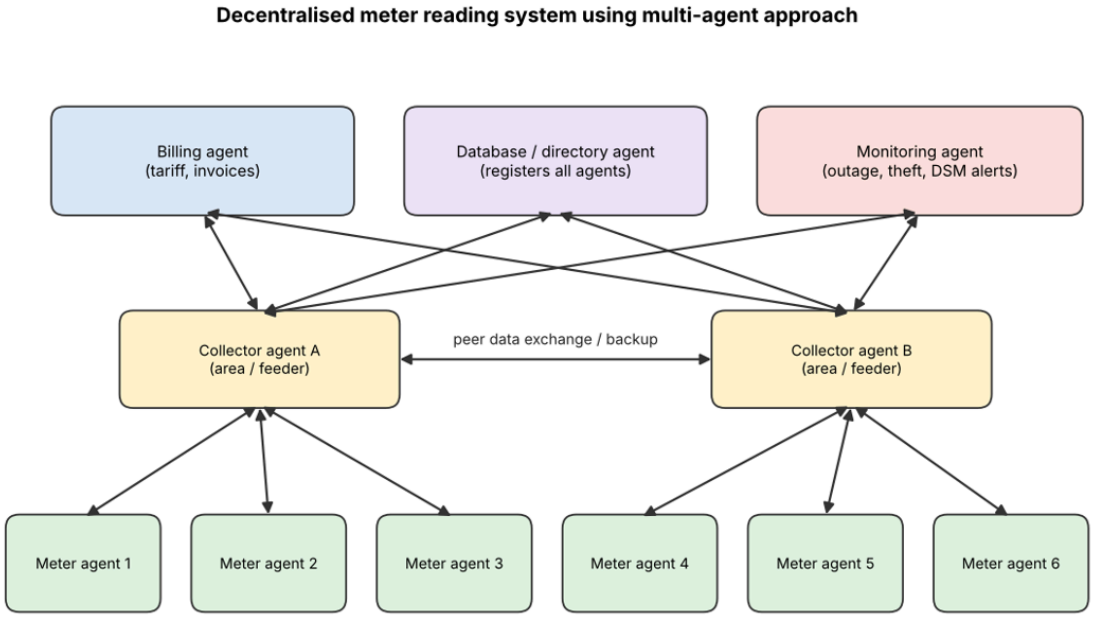
Intelligent distributed autonomous power system (IDAPS) with four agent types, drawn for this pack

IDAPS (intelligent distributed autonomous power system) is a microgrid concept (Virginia Tech, Rahman et al.) in which distributed energy resources and loads in a neighbourhood are managed by a multi-agent system so that the area can supply itself, trade energy and **island** during an upstream outage to keep critical loads supplied.

Agent	Function
Control agent	Monitors voltage and frequency at the point of connection; detects upstream faults/outages; decides to island the microgrid (opens the main breaker) and to reconnect when the grid is normal; sends signals to the other agents
DER agent	Stores information on each DER (type, rating, availability, cost); monitors and controls its output (start, stop, set-point); offers energy and bids in the local market
User (load) agent	Represents customers; monitors loads, knows which are critical; sheds or reschedules non-critical loads during islanding or high prices (demand response)
Database agent	Stores system data, agent directory and history; provides information to other agents

Operation: in normal conditions DER agents and user agents trade and schedule energy. When the control agent detects a fault upstream (e.g. voltage sag), it sends "island" messages: the main switch opens, the DER agents ramp up generation and storage, the user agent sheds non-critical loads so that critical loads remain supplied; when the grid recovers, the control agent re-synchronises and reconnects. The design steps: (1) define the system and objectives; (2) identify agents and their roles; (3) define their knowledge and behaviour (rules); (4) define message types and communication (FIPA-ACL, JADE platform); (5) implement and test by simulation.

Decentralised meter reading system using the multi-agent approach (2023 4d)



Decentralised meter reading system with agents drawn for this pack

1. **Meter agents** (one in each smart meter): measure and store interval consumption, detect tamper and outages, respond to read requests, and push data autonomously at scheduled times; can relay neighbours' messages (mesh).
2. **Collector agents** (one per feeder/area, in data concentrators): discover and register meter agents in their area, poll or receive readings, validate (check missing/abnormal values), aggregate, and forward to the upper level. Neighbouring collectors back each other up if one fails - no single point of failure.
3. **Directory/database agent**: keeps the registry of all agents and stores meter data (like an MDMS (meter data management system)).
4. **Billing agent**: applies tariffs (time-of-use, prepaid) to the data and generates bills; sends price signals back to meter agents.
5. **Monitoring agent**: analyses data for outages, theft (comparing feeder energy with sum of meters), voltage problems and demand-response events.
6. Agents communicate by standard messages (request, inform, agree, failure) over the NAN (neighbourhood area network)/WAN; new meters join by registering - plug and play.

Advantages: scalable to millions of meters, robust, reduced traffic (local aggregation), fast local decisions (outage detection), easy expansion.

> **Memory trick:** Agent = **"Autonomous, Reactive, Pro-active, Social"**. IDAPS agents = **"Control, DER, User, Database"**.

Section B: Power system control and stability, power quality, intelligence, security, renewables

B8 Computational intelligence and decision tools: ANN, expert systems, AHP, decision support tools, stability tools and computational challenges

Must study | Very likely · 100% | Easy

PAST-PAPER QUESTIONS

- **2023 Q7(b)** [10] What is meant by ANN (artificial neural network)? How does the ANN facilitate the assessment of smart grid performance?
- **2023 Q8(b)** [10] How do expert systems (ES) enhance decision-making and operational efficiency in smart grid? Discuss.
- **2022 Q7(b)** [10] How can decision support tools be used in smart grid planning and operations? Describe.
- **2022 Q7(c)** [10] What is the role of the Analytical Hierarchical Programming (AHP) method in smart grid management? Explain with necessary block diagram.
- **2022 Q8(d)** [12] Describe the computational challenges faced by researchers and engineers for the design, planning and operations of smart power system.
- **2020 Q5(a)** [10] What are the stability analysis tools for smart grid? Write the reasons of computational challenges in smart grids.
- **2020 Q5(c)** [10] How does an Artificial Neural Network (ANN) improve smart grid technology? Illustrate the Analytical Hierarchical Programming (AHP) method.

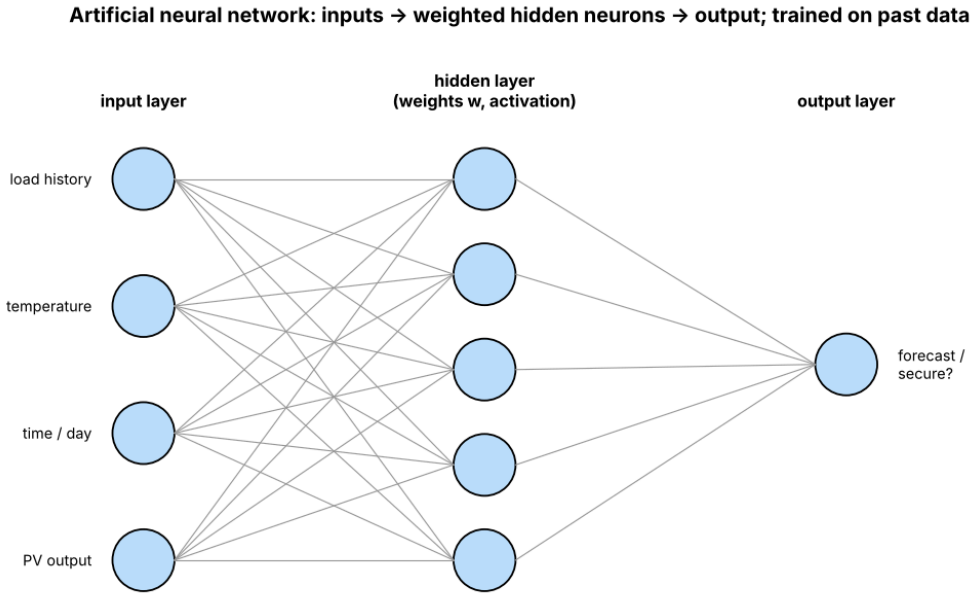
★ REMEMBER IT

- ANN = learns from past data (load and price forecasting, fault detection). Expert system = rules from human experts (if-then) for diagnosis and restoration. AHP (analytical hierarchy process (the paper calls it analytical hierarchical programming)) = compare options pair-wise against weighted criteria to rank them (choose a site, a technology).

Before you start: making the grid think

The smart grid produces huge amounts of data and must make decisions quickly under uncertainty. **Computational intelligence** tools help: **ANN** learns patterns from past data like a brain; **expert systems** store experts' rules (IF ... THEN ...); **AHP** helps choose between options by comparing criteria two at a time; **decision support tools** combine these with data and models to advise planners and operators.

ANN and how it facilitates assessment of smart grid performance (2023 7b, 2020 5c)



Artificial neural network (drawn for this pack)

An **artificial neural network (ANN)** is a computational model inspired by the brain, made of interconnected neurons arranged in layers (input, hidden, output). Each neuron forms a weighted sum of its inputs plus a bias and passes it through an activation function (sigmoid, ReLU). The network is **trained** on example data (e.g. back-propagation adjusts the weights to minimise error), after which it can map new inputs to outputs very fast, handle non-linear relationships and noisy data, and generalise.

Uses in the smart grid:

- **Load and price forecasting** (short-term, from weather, time, history).
- **Renewable forecasting** (solar irradiance, wind power).
- **Security and stability assessment**: classify operating states as secure/insecure quickly instead of running many load flows; voltage stability margin estimation.
- **Fault detection, classification and location** from waveforms; PQ disturbance classification.
- State estimation and bad-data/false-data detection; non-technical loss (theft) detection from meter data.
- Control: adaptive controllers for FACTS (flexible AC transmission systems), microgrids, EV charging, demand response.

Expert systems (2023 8b)

An **expert system (ES)** is a computer program that captures the knowledge of human experts as rules (IF-THEN) in a **knowledge base** and uses an **inference engine** (forward/backward chaining) with a working memory to reason about a situation and give advice with explanations through a user interface.

- **Alarm processing**: during a disturbance hundreds of alarms appear; the ES filters and identifies the root cause.
- **Fault diagnosis** and protection analysis from relay and breaker data.
- **Service restoration**: plans switching sequences quickly after outages (self-healing).
- Voltage/VAR control advice, security assessment, maintenance scheduling, operator training.

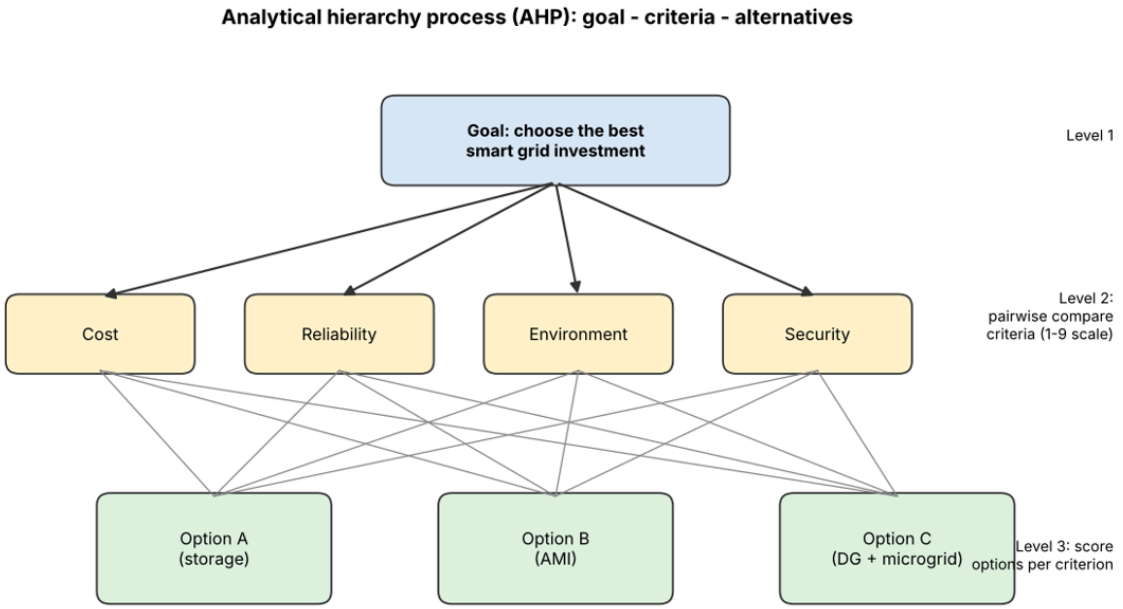
- Benefits: consistent fast decisions, stores scarce expertise, reduces operator errors and outage time, improves efficiency. Limits: knowledge acquisition is hard, cannot learn by itself (combine with ANN/fuzzy logic).

Decision support tools in planning and operations (2022 7b)

Decision support tools (DST) integrate data (SCADA, PMU (phasor measurement unit), AMI (advanced metering infrastructure), GIS (geographic information system), weather), models (load flow, OPF (optimal power flow), stability), optimisation and AI with visualisation to help humans decide.

- **Planning**: load and DG (distributed generation) forecasting, siting and sizing DG/storage, network expansion, reliability and cost-benefit studies, scenario analysis, multi-criteria choice (AHP).
- **Operations**: real-time visualisation (dashboards, GIS), contingency and security assessment, OPF/dispatch, Volt/VAR optimisation, outage management and restoration, demand response scheduling, alarm processing, what-if simulations.
- Benefits: faster and better-informed decisions, handling of uncertainty, reduced risk and cost.

AHP method (2022 7c, 2020 5c)



AHP hierarchy (drawn for this pack)

The **analytical hierarchy process (AHP)** (Saaty) is a multi-criteria decision-making method. Role in smart grid management: choosing among investment options (AMI vs storage vs microgrid), ranking DG sites, selecting communication technology or vendors, prioritising projects, evaluating smart grid maturity.

1. **Build the hierarchy**: goal at the top, criteria (cost, reliability, environment, security...) in the middle, alternatives at the bottom.
2. **Pairwise comparison** of criteria using Saaty's 1-9 scale (1 = equal, 9 = extremely more important) → comparison matrix A ($a_{ji} = 1/a_{ij}$).
3. **Priority weights** = normalised principal eigenvector of A (or average of normalised columns).
4. **Consistency check**: CI = $(\lambda_{max} - n)/(n - 1)$, CR = CI/RI; accept if CR < 0.1.
5. Repeat pairwise comparisons of alternatives under each criterion.

6. **Synthesis**: overall score of each alternative = Σ(criterion weight × alternative weight); highest score is chosen.

Stability analysis tools and computational challenges (2020 5a, 2022 8d)

Stability analysis tools: time-domain simulation (transient stability programs), equal area criterion, small-signal (eigenvalue/modal) analysis, P-V/Q-V curves and continuation power flow, voltage stability indices, direct (Lyapunov/energy function) methods, PMU-based real-time monitoring (WAMS), and AI-based security assessment (ANN, decision trees).

Computational challenges for design, planning and operation:

1. **Size and complexity**: millions of nodes (transmission + distribution + customer devices), many control loops.
2. **Uncertainty**: variable renewables, EVs, prices, customer behaviour → stochastic optimisation and many scenarios.
3. **Real-time requirements**: decisions in seconds with huge data streams from PMUs and AMI (big data, latency).
4. **Non-linear, multi-objective, mixed-integer optimisation** (OPF with switching, unit commitment, reconfiguration).
5. **Multi-time-scale dynamics**: power electronics (µs) to markets (hours), low inertia.
6. **Distributed control and communication** coordination; data quality, missing data and cyber attacks.
7. Modelling of new devices (inverters, storage, loads), lack of validated models; integration of different software and standards (interoperability).
8. Need for high-performance, parallel and cloud computing, and skilled people.

> **Memory trick: ANN learns, ES reasons by rules, AHP ranks by pairwise comparison, DST combines all for humans**. AHP steps: **"Hierarchy, Pairwise, Weights, Consistency, Synthesis"**.

B9 Interoperability and cyber security: threats, false data injection attacks, mitigation

Must study · Very likely · 100% · Easy

PAST-PAPER QUESTIONS

- **2023 Q7(c)** [13] What is interoperability in smart grid environment? How can false data injection impact the stability and reliability of smart grid operations, and what measures can be taken to mitigate these risks effectively?
- **2022 Q7(d)** [7] Explain why cyber security is of prime importance in smart grids and how the risk of cyber-attack can be mitigated.
- **2020 Q6(a)** [12] What is interoperability in smart grid environment? How is cyber security a threat to smart grids?
- **2020 Q6(b)** [12] What is false data injection attack in the smart grid? (AGC part in B2)
- **2020 Q6(c)** [6] What are the main mitigation approaches to cyber security risk?

UNDERSTAND **Read this to understand; you do not need to write it**

Interoperability = devices and systems from different makers can exchange data and work together (common standards: IEC (International Electrotechnical Commission) 61850, DNP3 (Distributed Network Protocol 3), IEC 61968/61970). **False data injection**: an attacker changes meter or PMU (phasor measurement unit) readings so that the state estimator believes a wrong grid state without raising a bad-data alarm; operators then dispatch wrongly, overload lines or shed load. Mitigation: encryption and authentication, secure PMUs at key buses, anomaly detection, network segmentation, firewalls, training, regular audits.

Before you start: plugging the grid into the internet

A smart grid connects millions of meters, sensors and controllers by networks. That brings two new problems: (1) devices from different makers must **understand each other** - interoperability; (2) every connected device is a door that a hacker might open - **cyber security**.

The three goals of cyber security are **CIA (confidentiality, integrity, availability)**: Confidentiality (no one reads what they should not), Integrity (no one changes data), Availability (the system keeps working). In the grid, availability and integrity matter most.

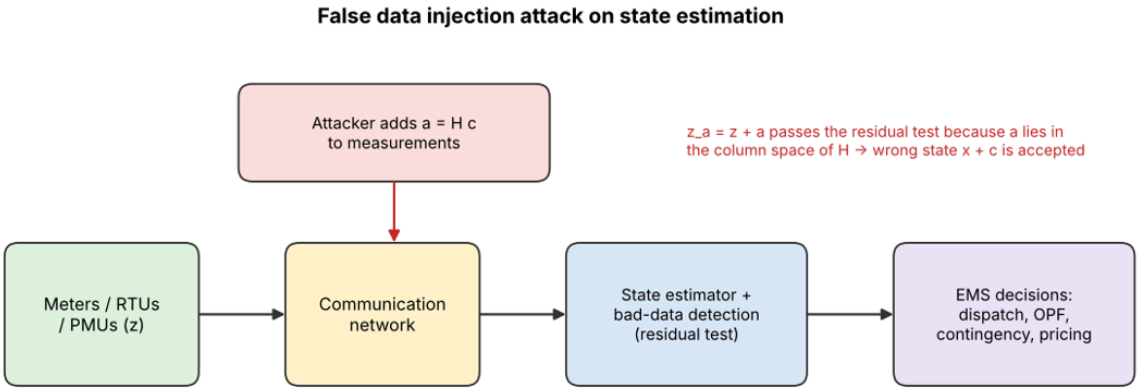
Interoperability (2023 7c, 2020 6a)

Interoperability is the capability of two or more networks, systems, devices, applications or components to exchange information securely and effectively and to use the information exchanged, with little or no inconvenience to the user (NIST/IEEE 2030). It requires common standards for communication protocols, data models and interfaces: IEC 61850 (substation automation), IEC 61968/61970 CIM (common information model), DNP3, IEC 60870-5, IEEE (Institute of Electrical and Electronics Engineers) C37.118 (PMU), ANSI (American National Standards Institute) C12 and DLMS (Device Language Message Specification)/COSEM (metering), OpenADR (demand response), IEEE 2030. Benefits: plug-and-play devices, vendor competition and lower cost, easier integration and upgrades, protection of investments, better security.

Why cyber security is of prime importance; how it threatens smart grids (2022 7d, 2020 6a)

- The grid is critical infrastructure: an attack can cause blackouts affecting hospitals, water, transport, economy and national security (e.g. Ukraine 2015/2016 grid attacks, Stuxnet).
- Huge attack surface: millions of smart meters, IEDs, DG (distributed generation) inverters, EV chargers and wireless links, many in unprotected public places.
- Two-way communication and remote control (connect/disconnect, set-points) can be misused to switch off loads or destabilise the grid.
- Legacy SCADA (supervisory control and data acquisition) protocols were designed without security; IT-OT convergence brings internet threats into control systems.
- Privacy: detailed meter data reveal customers' habits.
- Threats: malware/ransomware, denial-of-service (blocks AMI (advanced metering infrastructure)/SCADA communication), man-in-the-middle, spoofing, eavesdropping, meter tampering and energy theft, insider attacks, false data injection, GPS (Global Positioning System) spoofing of PMUs.

False data injection (FDI) attacks: impact and mitigation (2023 7c, 2020 6b)



False data injection attack on state estimation (drawn for this pack)

The control centre estimates the system state x from measurements z = Hx + e (H = measurement Jacobian) and uses a residual test to reject bad data. In a **false data injection attack**, an attacker who knows the network model adds a carefully designed vector a = Hc to the measurements; the estimated state becomes x + c but the residual is unchanged, so the **bad-data detector does not notice**.

Impact on stability and reliability: operators and EMS (energy management system) see a false picture of the grid; wrong economic dispatch and OPF (optimal power flow) decisions; hidden line overloads or fake congestion leading to tripping and cascading failures; wrong AGC (automatic generation control)/voltage control actions that may cause frequency or voltage instability; manipulated market prices for profit; masking of real faults or physical attacks; energy theft through meter data.

Mitigation: protect a strategically chosen set of key measurements (secured meters, encrypted channels) so that undetectable attack vectors cannot be built; deploy PMUs (independent, GPS-time-stamped measurements); authentication and digital signatures, encryption of measurement

data; advanced detection (statistical/consistency checks, machine-learning anomaly detection, comparing with forecasts); keep the network model (H) confidential; moving-target defence (changing system parameters); redundancy.

Main mitigation approaches to cyber security risk (2022 7d, 2020 6c)

1. **Risk assessment and security policy**: identify assets, threats and vulnerabilities; follow standards (NISTIR 7628, NERC (North American Electric Reliability Corporation) CIP (critical infrastructure protection), IEC 62351, IEC 62443).
2. **Defence in depth**: network segmentation, firewalls, DMZ (demilitarised zone (buffer network)) between IT and OT (operational technology), intrusion detection/prevention systems.
3. **Strong authentication and access control**: role-based access, multi-factor authentication, key management.
4. **Encryption and integrity protection** of communications (IEC 62351, TLS (Transport Layer Security), VPN (virtual private network)), secure protocols.
5. **Secure devices**: tamper-resistant meters, secure firmware updates, patch management, disabling unused services.
6. **Continuous monitoring**, logging, anomaly detection and a security operations centre.
7. **Incident response and recovery plans**, backups, drills, resilient (islandable) design.
8. **Training and awareness** of staff; supply-chain security; information sharing; privacy protection of customer data.

> **Memory trick:** Security goals **CIA** (in the grid: Availability and Integrity first). FDI (false data injection) = **"a = Hc fools the detector"**.

B3 Reactive power and voltage control: why the grid needs reactive power, control methods, AVR, SVC, improving P_{max} Must study Very likely · 100% Easy

PAST-PAPER QUESTIONS

- **2023 Q5(b)** [13] How can the maximum power output (P_{max}) of a generator be effectively improved? What are the primary functions of a thyristor-based SVC (static var compensator) in power system? Explain with diagram.
- **2023 Q6(c)** [15] What are the primary methods and devices used to control reactive power in power systems? How does the AVR (automatic voltage regulator) maintain voltage stability in smart grid? (loop part in B2)
- **2022 Q6(d)** [7] Why does the grid need reactive power? Discuss.
- **2020 Q4(c)** [8] What is the necessity of reactive power control in grid? Write the different methods of reactive power control.

★ REMEMBER IT

- Reactive power (Q) holds up voltage. Too little Q → low voltage, possible voltage collapse. Q sources: generators (AVR), shunt capacitors, synchronous condensers, SVC (static var compensator), STATCOM (static synchronous compensator), tap-changing transformers.
- P_{max} = EV/X: raise E (stronger excitation), raise V (voltage support), or lower X (parallel lines, series capacitors).

Before you start: active and reactive power in plain words

Active power P (watts) does useful work. Reactive power Q (var) does no net work but is needed to build the magnetic fields in motors, transformers and long lines; it also controls **voltage**. Rule of thumb: **P controls frequency and angle, Q controls voltage**.

Power through a line: P = (E V / X) sin δ, where E, V are the voltages at the ends, X the line reactance and δ the angle between them. The largest possible value (at δ = 90°) is **P_{max} = E V / X**.

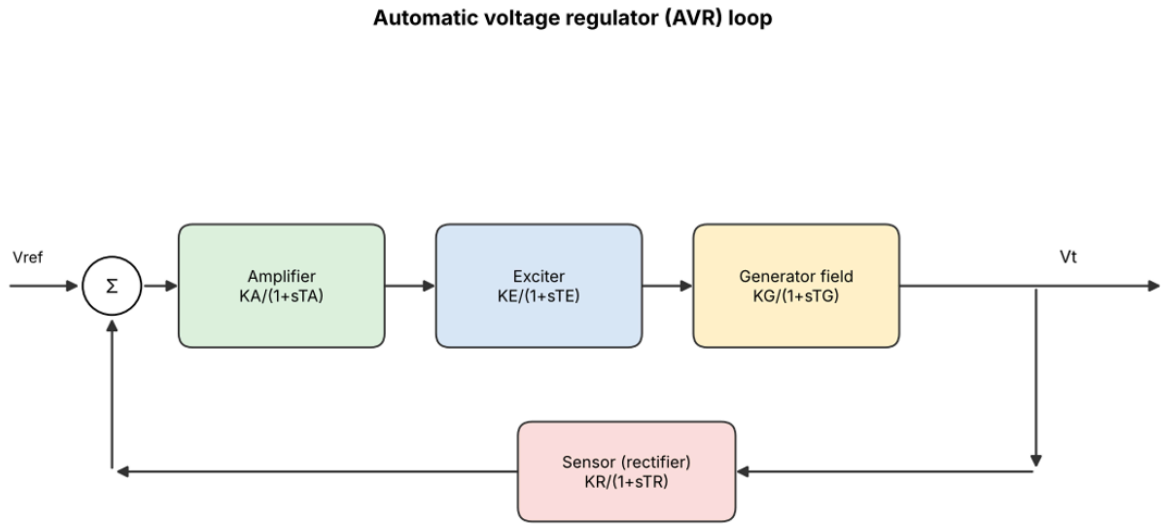
Why the grid needs reactive power (2022 6d, 2020 4c)

- Inductive loads (induction motors, transformers, fluorescent ballasts, fans, pumps, compressors) need Q for their magnetising fields; without it they cannot work.
- Lines and transformers absorb Q (I^2X); Q must be supplied to keep voltages within limits (±5-10 %). Lack of Q causes low voltage and, in the worst case, **voltage collapse** (B6).
- Reactive power supports power transfer: P_{max} = EV/X falls when voltages drop.
- Controlling Q reduces line current, so reduces I^2R losses and frees capacity; improves power factor (and avoids pf penalties).
- Inverter-based renewables must supply/absorb Q to avoid voltage rise/fall on feeders.

Methods and devices to control reactive power (2023 6c, 2020 4c)

Device / method	How it controls Q and voltage
Generator excitation with AVR (automatic voltage regulator)	Over-excited generator supplies Q, under-excited absorbs Q; the AVR holds terminal voltage
Shunt capacitors (fixed or switched)	Supply Q at load points, raise voltage, improve pf; cheap
Shunt reactors	Absorb Q on long lightly-loaded lines / cables (Ferranti effect) to limit over-voltage
Series capacitors	Cancel part of line reactance X: more power transfer, better voltage
Synchronous condensers	Synchronous motor running without load; continuously variable Q supply/absorption, adds inertia
Tap-changing transformers (OLTC) and regulating transformers	Adjust turns ratio to correct voltage (redistribute Q)
SVC (static var compensator)	Thyristor-controlled reactor + thyristor-switched capacitor: fast continuous Q control
STATCOM and other FACTS (UPFC, TCSC (thyristor-controlled series capacitor))	Voltage-source converter injects controllable Q; faster and better at low voltage than SVC
Inverter-based DG (distributed generation) **/ storage**	Smart inverters provide Volt/VAR support (Q droop) on distribution feeders
Load-side: pf correction, demand response, load shedding	Reduce Q demand in emergencies

How the AVR maintains voltage stability (2023 6c)



Automatic voltage regulator loop (drawn for this pack)

The **AVR** measures the generator terminal voltage V_t through a **sensor** (potential transformer + rectifier and filter), compares it with the reference V_{ref}; the error is amplified (**amplifier**) and drives the **exciter**, which changes the field current of the generator. If V_t falls (heavy load or reactive demand), the error increases the field current → internal emf E rises → generator supplies more Q → V_t returns to the set value. If V_t rises, excitation is reduced. With fast static exciters and a power system stabiliser (PSS) to damp oscillations, the AVR keeps voltages steady, maintains reactive

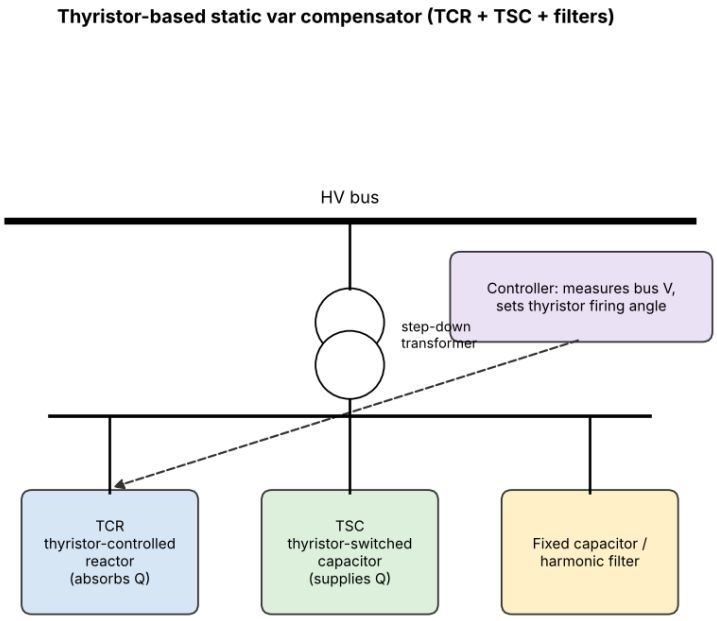
reserves and improves both voltage and transient stability. In the smart grid, AVR set-points can be coordinated with SVCs, OLTCs and DG inverters by Volt/VAR optimisation.

Improving P_{max} of a generator (2023 5b)

```
P_max = E V / X
```

1. **Increase E**: higher excitation (fast-acting exciters and AVR, high ceiling voltage).
2. **Increase V**: hold the system voltage up with reactive support (SVC, STATCOM, capacitors, synchronous condensers).
3. **Reduce X** (most effective): parallel (double-circuit) lines, bundled conductors, series capacitor compensation (TCSC), lower-reactance transformers and generators, higher transmission voltage.
4. Use FACTS (flexible AC transmission systems)/HVDC to control flows; also fast fault clearing and auto-reclosing to keep the post-fault P_{max} high.

Thyristor-based SVC: functions with diagram (2023 5b)



Static var compensator (drawn for this pack)

An **SVC** is a shunt-connected FACTS device made of a **thyristor-controlled reactor (TCR)** - a reactor in series with back-to-back thyristors whose firing angle (90°-180°) varies the effective inductance continuously - and **thyristor-switched capacitors (TSC)** switched in steps, plus fixed capacitors/harmonic filters, connected via a step-down transformer. A controller measures the bus voltage and sets the firing angles so the SVC behaves like a variable susceptance, from capacitive (supplying Q) to inductive (absorbing Q).

Functions: fast dynamic voltage control at the bus; reactive power compensation; improving transient stability and increasing power transfer (P_{max}); damping power oscillations; reducing voltage flicker (arc furnaces); balancing unbalanced loads; preventing voltage collapse.

> **Memory trick: P ↔ frequency, Q ↔ voltage**. P_{max} = EV/X: **raise E, raise V, cut X**. SVC = **TCR (absorbs) + TSC (supplies)**.

B11 Barriers and solutions, standards and research areas, the simplest and the most complex thing about the smart grid

Must study · Very likely · 100% · Easy

PAST-PAPER QUESTIONS

- **2023 Q8(a)** [13] What are the primary barriers to the development of smart grids, and what solutions can be implemented to overcome these obstacles? Discuss.
- **2022 Q8(c)** [8] Why are smart grid standards necessary? What would be the suitable areas of research in smart grid?
- **2020 Q4(a)** [12] What are the simplest thing and the complex thing about the smart grid? (voltage stability part in B6)

★ REMEMBER IT

- Barriers: high cost, unclear regulation, interoperability/standards, cyber security and privacy, skills shortage, customer acceptance. Solutions: pilot projects, standards, incentives, training, public awareness.

Before you start: why smart grids are slow to come

The idea is simple, but turning a century-old grid into a smart one costs a lot, needs new rules, new skills and common standards, and must be safe against hackers. This topic lists those barriers, their solutions, the role of standards and the open research areas.

The simplest and the most complex thing about the smart grid (2020 4a)

Simplest: the concept itself - add sensing, two-way communication and computer control to the existing power grid so that it can monitor itself and respond automatically ("an internet for electricity"). Many pieces (smart meters, SCADA (supervisory control and data acquisition), automation) already exist and can be added step by step.

Most complex: making it all work together - integrating millions of heterogeneous devices, variable renewables, storage and EVs, markets and active customers in real time, with interoperability, stability, data management and cyber security, under uncertain regulation and economics. The power system is the largest machine ever built; changing it while it runs is a huge systems-engineering challenge.

Primary barriers and solutions (2023 8a)

Barrier	Solution
High capital cost, uncertain payback, utilities' limited finance	Phased deployment starting with high-benefit projects (prepaid/smart meters, loss reduction); cost-benefit studies; government funds, green bonds, development banks (ADB, World Bank), public-private partnerships
Regulatory and policy barriers: tariffs that reward selling more kWh, no time-of-use pricing, unclear cost recovery	Smart grid roadmap and policy; regulator (BERC) to allow cost recovery, performance-based regulation, dynamic tariffs, net metering
Lack of standards and interoperability; legacy equipment	Adopt international standards (IEC 61850, CIM (common information model), IEEE (Institute of Electrical and Electronics Engineers) 2030); open architecture; pilot testing
Cyber security and privacy concerns	Security by design, standards (NISTIR 7628, IEC (International Electrotechnical Commission) 62351), data protection law, monitoring
Technical complexity: integrating renewables, stability, communication coverage	Research, pilots and demonstration projects; storage; advanced control and AI
Skills shortage	Education and training programmes, university-industry collaboration
Consumer acceptance: privacy fears, distrust of new meters and bills, low awareness	Awareness campaigns, transparent billing, apps, incentives, protecting data
Institutional inertia, many stakeholders with different interests	Strong leadership, stakeholder coordination, clear roles

Why smart grid standards are necessary (2022 8c)

- **Interoperability**: devices and systems from different vendors work together (plug and play).
- **Reduce cost and risk**: competition among vendors, avoid lock-in and stranded assets, economies of scale.
- **Security and safety**: common cyber security and electrical safety requirements.
- **Reliability and quality**: performance requirements (e.g. PMU (phasor measurement unit) accuracy IEEE C37.118, DG (distributed generation) interconnection IEEE 1547, PQ IEEE 519).
- **Scalability and future-proofing**: allow upgrades and new applications.
- Facilitate regulation, certification, international trade and knowledge sharing.

Suitable areas of research in smart grid (2022 8c)

- Renewable and load forecasting; integration of high renewable shares (grid-forming inverters, synthetic inertia).
- Energy storage technologies and their optimal operation; EVs and V2G (vehicle-to-grid).
- Microgrids, nanogrids, virtual power plants, peer-to-peer energy trading (blockchain).
- Wide-area monitoring, protection and control with PMUs; real-time state estimation.
- Cyber security (false data injection detection), privacy-preserving analytics.
- AI and big data for operation, predictive maintenance and theft detection; DSOPF (dynamic stochastic optimal power flow) and optimisation under uncertainty.
- Demand response and customer behaviour; dynamic pricing and market design.
- Communication networks (5G, IoT (Internet of Things)) for the grid; interoperability standards.

- Power quality with power electronics; adaptive protection in active networks; resilience to climate disasters.

> **Memory trick:** Barriers = **"Cost, Rules, Standards, Security, Skills, Society"**.

B10 Renewables and the smart grid: integration challenges, education and training, V2G, sustainable energy options, efficiency and DG control

Must study | Very likely · 83% | Easy

PAST-PAPER QUESTIONS

- **2023 Q7(a)** [12] What are some current research challenges in integrating renewable energy sources into the smart grid infrastructure? How does education and training contribute to the successful implementation of smart grid?
- **2023 Q6(b)** [8] Write a short note on vehicle-to-grid (V2G) technology.
- **2022 Q8(a)** [8] How does a smart grid improve efficiency? What are the benefits of distributed generation control in smart grid?
- **2022 Q8(b)** [7] Give an overview of sustainable energy options for the smart grid.

★ REMEMBER IT

- V2G (vehicle-to-grid): parked EVs plugged in can send stored energy back to the grid at peak times and charge at night: a huge distributed battery for frequency and peak support.

Before you start: renewables, EVs and efficiency

Solar and wind are clean but **variable** (clouds, wind gusts) and **uncertain**; they connect through inverters (no spinning mass, so less inertia) and are spread over many small sites. The smart grid gives the tools to integrate them: forecasting, storage, flexible demand and fast control.

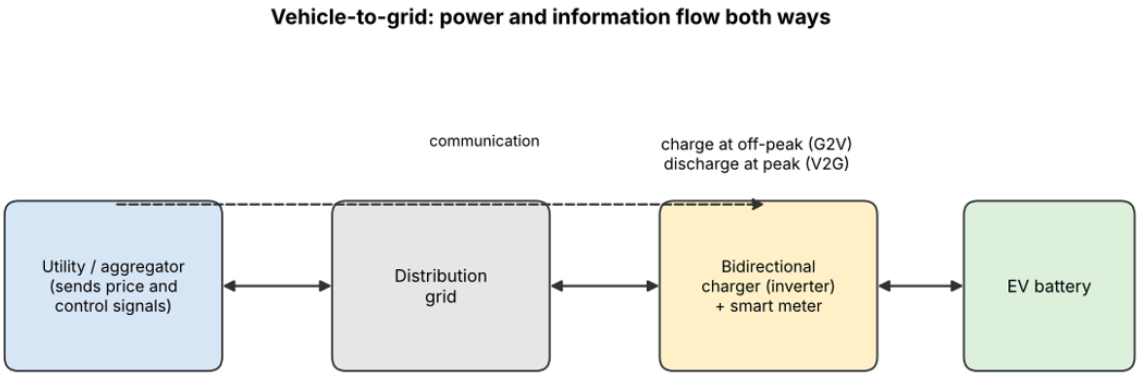
An electric vehicle is a big battery on wheels that is parked about 95 % of the time. **V2G (vehicle-to-grid)** (vehicle-to-grid) lets it give power back to the grid when needed.

Research challenges in integrating renewables (2023 7a)

1. **Variability and forecasting**: accurate short-term solar/wind forecasts; probabilistic planning and operation.
2. **Low inertia and stability**: frequency and voltage stability with high inverter share; grid-forming inverters, synthetic inertia, fast frequency response.
3. **Energy storage**: low-cost, long-life storage, optimal sizing and control.
4. **Distribution impacts**: voltage rise, reverse power flow, hosting capacity, protection coordination and islanding detection.
5. **Power quality**: harmonics and flicker from inverters.
6. **Control and optimisation**: coordinated control of millions of DERs, microgrids, virtual power plants, demand response.
7. **Market and regulatory design**: pricing flexibility and ancillary services, net metering.
8. **Communication, data and cyber security** for distributed resources; interoperability standards.

Role of education and training: smart grids need engineers and technicians skilled in power systems, power electronics, ICT (information and communication technology), data science and cyber security together. Education and training (university courses, vocational programmes, utility staff training, simulators, certification) build this workforce; they help operators use new tools (DMS, WAMS (wide area monitoring system)), reduce errors, support research and innovation, and inform consumers so they participate in demand response, rooftop PV and EVs. Without trained people, smart grid investments fail.

Vehicle-to-grid (V2G) technology (2023 6b)



Vehicle-to-grid concept (drawn for this pack)

V2G is a system in which plug-in electric vehicles communicate with the grid and, through a **bidirectional charger**, can both draw power (G2V, grid-to-vehicle) and return stored energy to the grid (V2G), controlled by the utility or an aggregator according to price and grid needs.

- **Services**: peak shaving (discharge at evening peak), valley filling (charge at night), frequency regulation and spinning reserve (fast response), voltage support, storing surplus solar/wind, emergency backup (vehicle-to-home, V2H (vehicle-to-home)).
- **Benefits**: cheaper grid operation, better renewable integration, extra income for EV owners, less need for peaking plants.
- **Challenges**: battery degradation from extra cycling, cost of bidirectional chargers, communication and standards (ISO 15118), aggregation of many small vehicles, owner availability and willingness, cyber security.

How a smart grid improves efficiency; benefits of DG (distributed generation) control (2022 8a)

- **Loss reduction**: Volt/VAR optimisation, conservation voltage reduction, network reconfiguration, power factor correction, DG close to loads.
- **Peak reduction and load-factor improvement** by DSM (demand side management) and storage: fewer inefficient peaking plants, better use of assets.
- **Better asset utilisation**: dynamic line rating, condition-based maintenance, fewer outages.
- **Informed consumers**: real-time data and time-of-use prices reduce waste (typically 5-15 % savings).
- **Integration of efficient and clean generation** (CHP, renewables).

Benefits of distributed generation control: keeps feeder voltages within limits (reactive power control of inverters); prevents reverse-flow overloads; enables DG to provide ancillary services (frequency response, reserves); coordinated dispatch minimises losses and cost; supports islanding and microgrid operation for reliability; increases DG hosting capacity; smooths renewable output with storage; allows virtual power plants to take part in markets.

Sustainable energy options for the smart grid (2022 8b)

Option	Role
Solar PV (rooftop, solar parks), solar thermal	Largest growth in Bangladesh; distributed generation with smart inverters
Wind (onshore, offshore)	Bulk renewable energy; needs forecasting and balancing
Hydro and pumped hydro	Flexible renewable generation and large-scale storage (Kaptai)
Biomass, biogas, waste-to-energy	Dispatchable renewable DG, rural mini-grids
Geothermal, tidal/wave	Location-specific renewables
Fuel cells and green hydrogen	Clean DG, long-term storage, CHP (combined heat and power)
Energy storage (batteries, flywheels, supercapacitors)	Firm up renewables, ancillary services
Electric vehicles and V2G	Flexible load and mobile storage
Energy efficiency, CHP, demand response	The cheapest "resource": reduce demand and losses
Nuclear (low carbon)	Base-load low-carbon supply (Rooppur)

Memory trick: Renewable challenges = **"Variable, Uncertain, Low-inertia, Distributed"**. V2G = **charge at night, give back at peak, regulate frequency**.

B7 Power quality: issues in a smart grid, importance of power-quality management and audit

Must study | Very likely · 83% | Easy

PAST-PAPER QUESTIONS

- **2023 Q5(c)** [7] Why is power quality management in smart grid so important?
- **2022 Q5(b)** [8] What are the power quality issues with smart grid? What is the importance of power quality audit in smart grid? Discuss.

★ REMEMBER IT

- Power quality problems: **sags/swells, harmonics, flicker, transients, interruptions, unbalance, frequency deviation**. Smart grids have more of them because of inverters (harmonics) and variable renewables (fluctuations).

Before you start: clean power

Ideal electricity is a perfect 50 Hz sine wave at exactly rated voltage, never interrupted. Anything that distorts or disturbs this is a **power quality (PQ)** problem. Computers, drives, PLCs and medical equipment are very sensitive to it, and the smart grid adds many power-electronic sources (inverters) that can create it.

Power quality issues in a smart grid (2022 5b)

Issue	Meaning	Typical cause
Voltage sag (dip)	Short drop to 10-90 % for 0.5 cycle-1 min	Faults, motor starting
Swell	Short rise above 110 %	Single-line faults, load switching off, DG (distributed generation)
Interruption	Voltage < 10 % (momentary to sustained)	Faults, breaker operation
Transients / surges	Very fast spikes	Lightning, capacitor switching
Harmonics	Waveform distortion at multiples of 50 Hz (THD)	Inverters, VFDs, LEDs, SMPS (switched-mode power supply), EV chargers
Flicker / voltage fluctuation	Repeated small variations visible in lamps	Arc furnaces, wind/PV output variation
Unbalance	Unequal three-phase voltages	Single-phase loads and rooftop PV
Frequency deviation	Departure from 50 Hz	Generation-load imbalance, low inertia
DC offset, notching, noise		

Importance of power quality management in the smart grid (2023 5c)

- Sensitive digital loads (computers, PLCs, medical, telecom, data centres) malfunction or fail; industrial process interruptions cost a lot.
- Harmonics cause overheating of transformers, cables and motors, extra losses, capacitor failure, nuisance tripping and metering errors.

- High penetration of inverter-based renewables, EV chargers and LEDs increases harmonics, voltage rise and flicker - must be managed.
- Smart grid communication and sensing equipment itself needs clean power.
- Better PQ = higher efficiency, longer equipment life, customer satisfaction and compliance with standards (IEEE 519, IEC (International Electrotechnical Commission) 61000, EN 50160).
- Smart meters and PMUs make continuous PQ monitoring possible, and FACTS (flexible AC transmission systems)/custom power devices (DVR, STATCOM (static synchronous compensator), active filters, UPS (uninterruptible power supply)) fix problems.

Importance of power quality audit (2022 5b)

A **PQ audit** is a systematic measurement and analysis of voltage, current, harmonics, sags, flicker, pf and unbalance at a site (with PQ analysers over a period), comparing with standards, finding sources and recommending solutions.

1. Identifies PQ problems and their sources before equipment is damaged.
2. Quantifies losses (harmonic and reactive) and energy-saving potential; pf correction.
3. Guides the right mitigation: filters, UPS, DVR (dynamic voltage restorer), isolation transformers, wiring/grounding fixes.
4. Verifies compliance with grid codes and standards (important for DG/PV connection).
5. Gives a baseline for planning and for contracts between utility and customer.
6. Improves reliability and reduces downtime and maintenance costs.

> **Memory trick:** PQ problems = **"Sag, Swell, Stop (interruption), Spike, Harmonics, Flicker, Unbalance"**.

B2 Frequency control: why grid frequency matters, turbine speed governing system, load frequency control loop, AGC

Must study · Very likely · 100% · Medium

PAST-PAPER QUESTIONS

- **2023 Q6(c)** [15] Draw and discuss the simple feedback loop block diagram of generator and load. What are the primary methods and devices used to control reactive power in power systems? And how does the AVR (automatic voltage regulator) maintain the voltage stability in smart grid? (reactive power part in B3)
- **2022 Q5(c)** [12] Why is power grid frequency important? How does the turbine speed governing system work? Explain with schematic diagram.
- **2022 Q5(d)** [9] What is Automatic Generation Control (AGC) in power system? "AGC (automatic generation control) is essential in smart grid" - justify.
- **2020 Q6(b)** [12] What is false data injection attack in the smart grid? "Automatic Generation Control (AGC) is essential for power system" - justify. (attack part in B9)

UNDERSTAND **Read this to understand; you do not need to write it**

Frequency shows the balance of generation and load in real time. More load than generation → generators slow down → frequency falls. The **governor** (primary control) senses speed and opens the steam/water valve within seconds, but leaves a small steady error set by the droop R. **AGC** (automatic generation control, secondary control) then slowly resets the governor set-points so frequency returns exactly to 50 Hz and tie-line flows to schedule.

CALCULATOR **Governor droop: frequency drop after a load increase**

Change any number; every step is recalculated and shown with the numbers put in.

1. Area frequency response: $\beta = 1/R + D = 1 /$ **0.05** + **1** = **21.00** pu/pu
2. Steady frequency deviation (pu): $\Delta f = -\Delta P_L/\beta =$ **-0.1** / **21** = **-0.00476** pu
3. Frequency drop: **(-0.004762)** × **50** = **-0.238** Hz
4. New frequency (no AGC): **50** + **(-0.2381)** = **49.762** Hz

AGC adds an integral controller that pushes Δf back to zero.

Before you start: why frequency tells you the balance

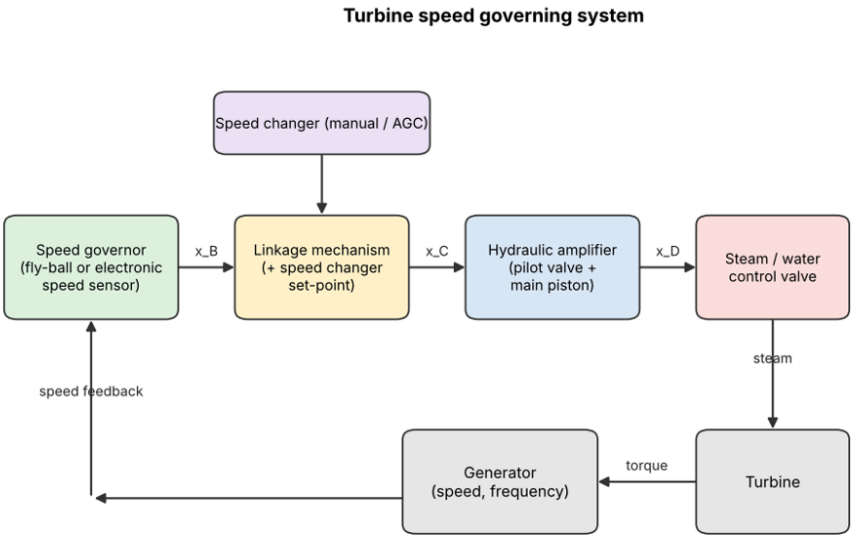
All generators in a grid spin in step (synchronously). Frequency (50 Hz in Bangladesh) is proportional to their speed. If load suddenly rises and generation does not, the extra energy is taken from the spinning rotors, they slow down and frequency falls. If generation exceeds load, frequency rises. So **frequency is the live indicator of the generation-load balance**.

Control happens in layers: inertia (instant), **primary control** by each machine's speed governor (seconds; frequency settles a little below 50 Hz), **secondary control = AGC** (minutes; brings frequency exactly back to 50 Hz), then tertiary/economic dispatch.

Why grid frequency is important (2022 5c)

- It shows the real-power balance; a big deviation means a generation-load mismatch that can lead to under-frequency load shedding or a blackout.
- Motors' speed and many industrial processes depend on frequency; clocks and timing devices use it.
- Generators and steam turbine blades can be damaged by operation away from rated frequency (blade resonance); transformers and motors overheat at low frequency (higher flux).
- Protection relays trip units outside the allowed band (e.g. 49.0-51.0 Hz), which can cascade.
- Interconnected areas must hold scheduled frequency to keep tie-line flows correct.

Turbine speed governing system (2022 5c)

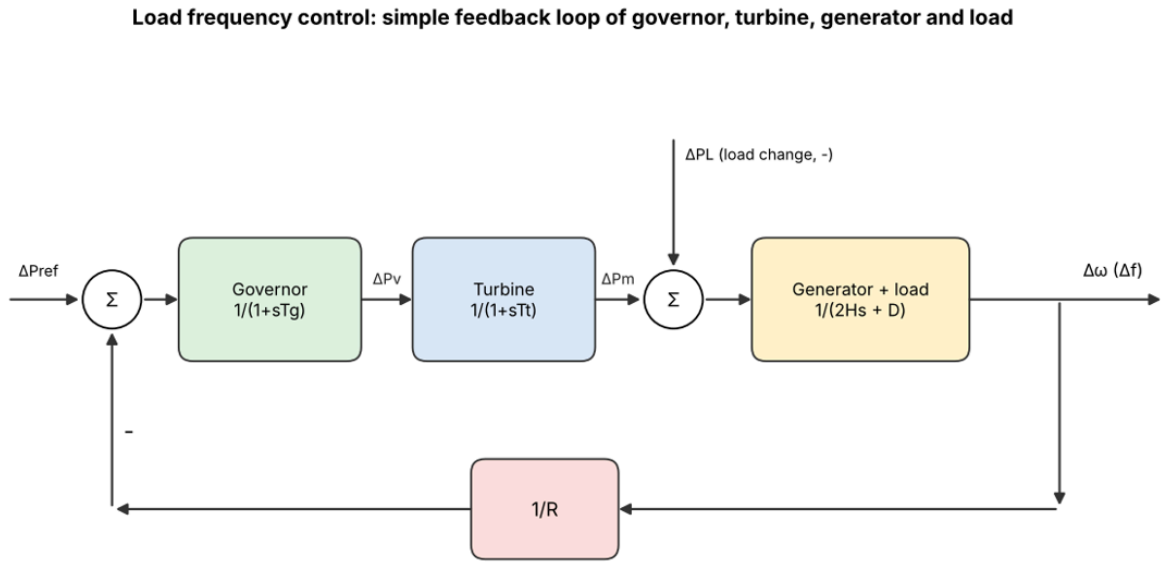


Turbine speed governing system (schematic, drawn for this pack)

1. **Speed governor** (fly-ball or electronic sensor) driven from the turbine shaft senses speed. When speed falls, the fly-balls move in and point B of the linkage moves.
2. **Linkage mechanism** converts the fly-ball movement into a displacement of the pilot valve (point C); it also gives feedback from the main piston so the valve settles at a new position (this gives the **droop** R).
3. **Speed changer** (load reference set-point, motor-driven): moved manually or by AGC to change the output at a given frequency.
4. **Hydraulic amplifier**: the small pilot-valve movement admits high-pressure oil to the main piston, which produces the large force needed to move the steam valve (point D).
5. **Steam (or water) control valve** opens more, turbine power rises, generator speed and frequency recover; the final frequency is slightly lower than before (steady-state error = R × ΔP).

```
Steady-state: Δf = -R ΔP_L / (1 + DR)   (R = droop, Hz/MW or pu; 5 % droop typical)
```

Simple feedback loop block diagram of generator and load (2023 6c)



Load frequency control loop (drawn for this pack)

The loop: a change in reference ΔP_{ref} minus the speed feedback $\Delta\omega/R$ drives the **governor** (time constant $T_g \approx 0.2$ s), which moves the valve ΔP_V; the **turbine** ($T_t \approx 0.5$ s) changes mechanical power ΔP_m; $\Delta P_m - \Delta P_L$ (load change) acts on the **generator-load** block 1/(2Hs + D) (H = inertia constant, D = load damping) giving the speed/frequency deviation $\Delta\omega$, which is fed back through the droop 1/R. This primary loop limits the frequency drop; a steady error remains unless AGC adds an integral loop that changes ΔP_{ref}.

Automatic generation control (AGC) and why it is essential in the smart grid (2022 5d, 2020 6b)

AGC is the secondary control system that automatically adjusts the set-points (speed changers) of selected generators to (1) restore system frequency to its scheduled value (50 Hz) after primary control, (2) keep tie-line power flows between control areas at their scheduled values, and (3) share load among generators economically. It works on the **area control error**: ACE (area control error) = $\Delta P_{tie} + B\Delta f$ (B = frequency bias); an integral controller drives ACE to zero every few seconds.

"AGC is essential" - justification:

1. Primary control alone leaves a permanent frequency error; only AGC removes it.
2. Keeps interchange with neighbours at contract values (important for Bangladesh-India links).
3. Renewables (solar, wind) cause fast, random power fluctuations and reduce inertia; AGC continuously balances them, helped by fast storage and demand response in the smart grid.
4. Economic: re-allocates load to the cheapest units (with economic dispatch).
5. Prevents under-frequency load shedding and cascading blackouts, so improves reliability.
6. Smart grid AGC uses PMU (phasor measurement unit) data, storage, EVs (V2G) and responsive loads as regulation resources.

> **Memory trick:** Frequency = **balance meter**. Layers: **inertia (instant) → governor/droop (seconds) → AGC/integral of ACE (minutes) → dispatch**. Governor chain: **"Fly-balls, Linkage, Hydraulic amplifier, Valve"**.

B6 Voltage stability: why instability occurs, preventive control, assessment techniques (P-V, Q-V, VIPI, indices) Must study Very likely · 100% Medium

PAST-PAPER QUESTIONS

- **2023 Q8(c)** [12] Mention the names of different voltage stability assessment techniques. Describe one of them.
- **2022 Q7(a)** [8] Why does voltage instability occur in power grid? Discuss the voltage instability preventive control scheme for the power grid.
- **2020 Q4(a)** [12] What are the simplest thing and the complex thing about the smart grid? Write about the major reasons for voltage stability problems in power system. (first part in B11)
- **2020 Q5(b)** [10] Classify the techniques of voltage stability assessment. Describe the VIPI (voltage instability proximity index) method for voltage stability assessment.

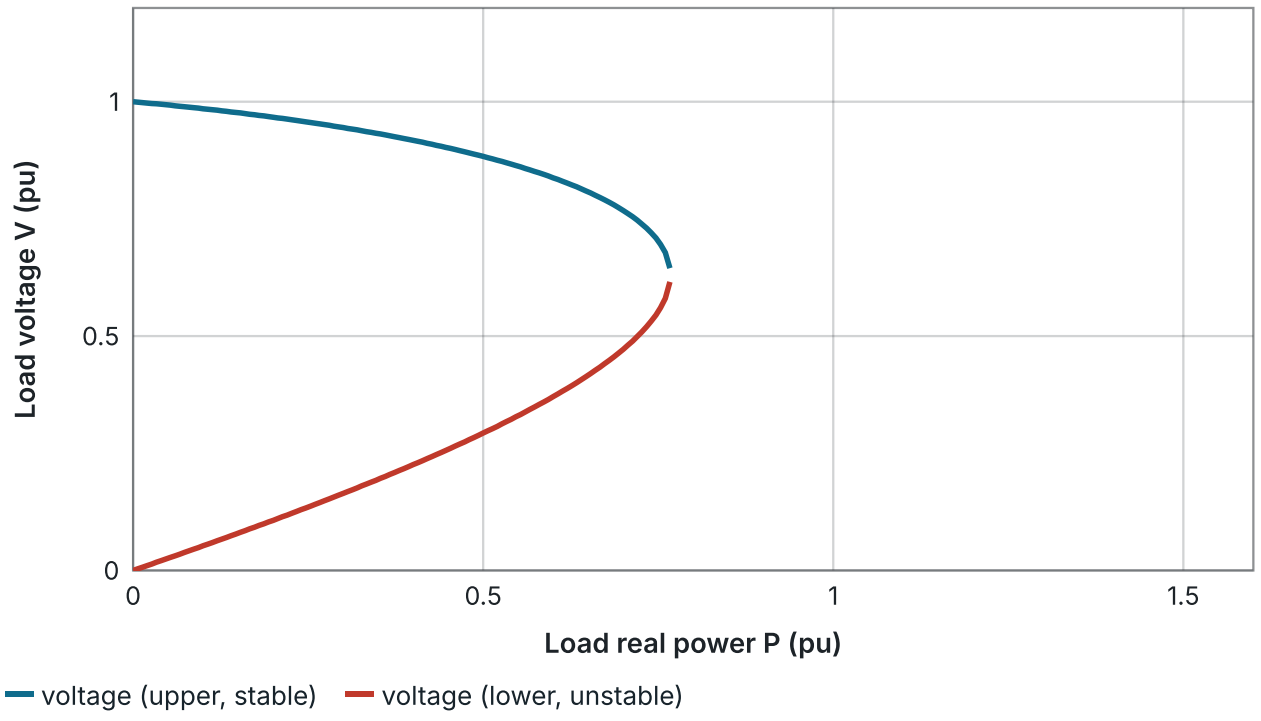
UNDERSTAND **Read this to understand; you do not need to write it**

Voltage instability happens when the network cannot deliver the reactive power the loads need: heavy loading, long lines, generators hitting their Q limits, or a line trip. Voltage then sags, motor loads draw more current, voltage sags more, until it collapses.

P-V curve ("nose curve"): as load P rises, voltage falls slowly then suddenly; the tip of the nose is the maximum loadability, and the distance from today's load to the nose is the **stability margin**.

INTERACTIVE **P-V nose curve of a simple line**

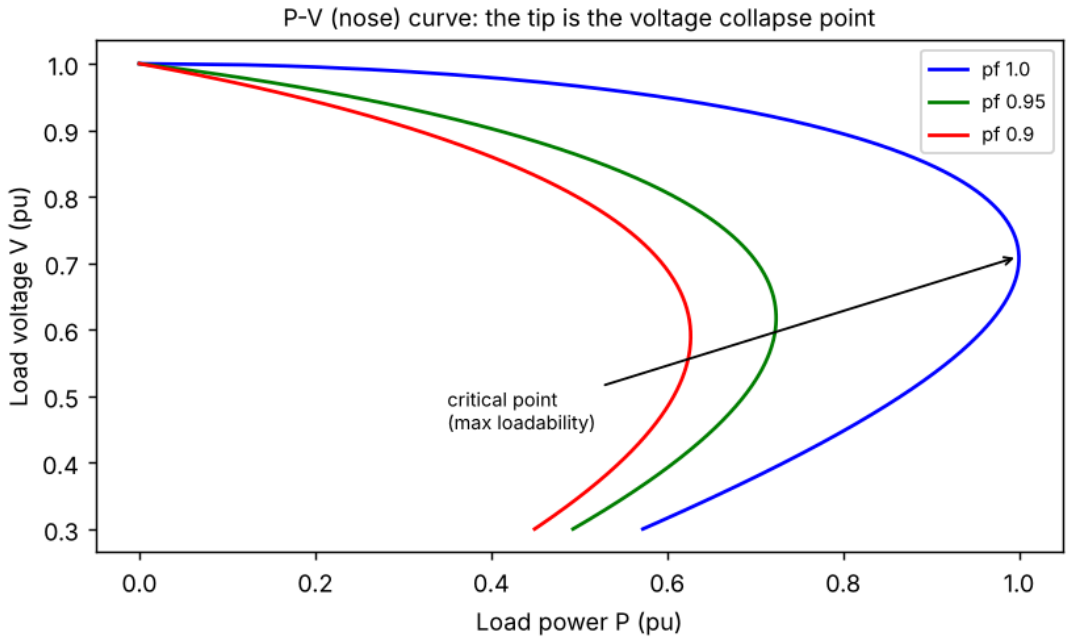
Move the sliders and watch the curve change.



The two branches meet at the nose (maximum power). A lagging load (bigger angle) moves the nose left; capacitive support moves it right.

Before you start: voltage collapse
As you draw more and more power through a line, the receiving-end voltage falls slowly at first, then faster, until a point where drawing more makes the voltage crash - the tip of the **P-V "nose" curve**. Beyond that tip no steady operating point exists: **voltage collapse** (blackout). The root cause is shortage of **reactive power** at the load end (Q controls voltage).

Why voltage instability occurs (2022 7a, 2020 4a)



P-V (nose) curve: the tip is the maximum loadability / collapse point (drawn for this pack)

Voltage instability is the inability of a power system to maintain acceptable voltages at all buses after a disturbance or load increase; voltages fall progressively and uncontrollably. Major reasons:

1. **Heavy loading** and long transmission distances (large reactance → large reactive losses I^2X).
2. **Shortage of reactive power**: generators reaching field-current (Q) limits, capacitor banks whose output falls with V^2.
3. **Outage of lines or generators** (contingencies) that weaken the network.
4. **On-load tap changers** restoring load-side voltage, which increases current and reactive demand on the weak transmission system.
5. **Load characteristics**: constant-power loads, induction motors that stall at low voltage and draw high reactive current, air-conditioners.
6. Poor coordination of controls and protection; low power factor loads; high penetration of inverter-based DG (distributed generation) without voltage support.

Voltage instability preventive control scheme (2022 7a)

- Keep adequate **reactive power reserves** on generators, SVCs and STATCOMs; switch in shunt capacitors, switch out reactors.

- Raise generator voltage set-points (AVR) and coordinate secondary voltage control.
- **Block or reverse OLTC** (on-load tap changer) action during emergencies.
- **Re-dispatch generation** closer to load centres; reduce transfers on weak corridors.
- Demand response and **under-voltage load shedding** as the last resort.
- On-line voltage stability assessment with PMUs, stability margins (distance to the nose), alarms and wide-area special protection schemes.

Voltage stability assessment techniques (2023 8c, 2020 5b)

Class	Techniques
Static (steady-state) methods	P-V curve analysis; Q-V curve analysis (reactive margin); continuation power flow (CPF) to trace the nose; modal (eigenvalue) analysis of the reduced Jacobian; singular value decomposition; minimum singular value of Jacobian
Voltage stability indices	L-index (Kessel-Glavitsch), line indices (L_{mn}, FVSI, LQP), VIPI (voltage instability proximity index), tangent vector index, sensitivity dV/dQ
Dynamic methods	Time-domain simulation including OLTCs, loads and limiters; small-signal analysis
Measurement-based / intelligent methods	Thevenin equivalent from PMU (phasor measurement unit) data (impedance matching), ANN (artificial neural network), decision trees, expert systems

Described method: VIPI (2020 5b)

The **voltage instability proximity index (VIPI)** (Tamura) uses the fact that a load-flow problem has **two solutions** for every loading below the critical point: the normal high-voltage solution and an abnormal low-voltage solution. As the load increases these two solutions move towards each other and **coincide at the critical (nose) point**.

1. Solve the load flow for the present loading and obtain the high-voltage solution vector x_1 and the low-voltage (multiple) solution x_2.
2. Form two vectors in the space of the specified quantities: from the origin to the specified load vector y_s, and to the "singular" vector $y(x_c)$ at the midpoint $x_c = (x_1 + x_2)/2$.
3. VIPI = the angle θ between these vectors: $\cos\theta = (y_s \cdot y(x_c)) / (|y_s||y(x_c)|)$.
4. Far from collapse the solutions are far apart and θ is large; as loading approaches the critical point $\theta \to 0$.
5. So VIPI gives a scalar measure of how close the system is to voltage collapse; operators act (preventive control) when it falls below a threshold.

Advantages: one number for the whole system, uses load-flow results; disadvantage: finding the low-voltage solution needs extra computation. (Alternatively describe the P-V curve method: increase load step by step, run load flow, plot V against P; the distance from the operating point to the nose is the MW margin.)

> **Memory trick:** Voltage collapse = **"not enough Q at the end of a long, heavy line"**. Methods: **"PV, QV, CPF** (continuation power flow)**, Modal, Indices (L, VIPI), Dynamic, PMU/AI"**.

B1 Buses and bus types, infinite bus, bus admittance matrix Y_{bus} and proof of $I_{bus} = Y_{bus}V_{bus}$

Must study · Very likely · 83% · Medium

PAST-PAPER QUESTIONS

- **2023 Q5(a)** [15] What are the bus and infinite bus in power system? Describe a method to derive the bus admittance matrix in smart power system and prove that $I_{bus} = Y_{bus}V_{bus}$.
- **2022 Q5(a)** [6] How many buses are there in the power system? What is the significance of different types of buses in the power system?

LEARN IT FROM ZERO

Bus admittance matrix, from zero

Write Kirchhoff's current law at every bus: the current injected at bus i equals the sum of currents leaving through all the lines connected to it. With line admittance $y_{ij} = 1/z_{ij}$, current from i to j is $y_{ij}(V_i - V_j)$. Collect the terms of each V and you get $\mathbf{I = Y_{bus}V}$.

The rules fall out automatically: **diagonal** Y_{ii} = sum of every admittance connected to bus i (including shunt to ground); **off-diagonal** $Y_{ij} = -y_{ij}$ (minus the line between i and j, zero if no line). Y_{bus} is symmetric and mostly zeros (sparse).

CALCULATOR

3-bus Y_{bus} from line reactances (lossless lines, values in pu)

Change any number; every step is recalculated and shown with the numbers put in.

1. Line 1-2 susceptance: y = 1/(jx) = −j/x, magnitude = 1 / **0.2** = **5.000** pu
2. Line 1-3 susceptance: 1 / **0.25** = **4.000** pu
3. Line 2-3 susceptance: 1 / **0.4** = **2.500** pu
4. Y_{11} (times −j): $Y_{11} = y_{12} + y_{13} + y_{10}$ = **5** + **4** - **0** = **9.000** pu
5. Y_{22} (times −j): **5** + **2.5** = **7.500** pu
6. Y_{33} (times −j): **4** + **2.5** = **6.500** pu
7. $Y_{12} = Y_{21}$ (times −j): $Y_{12} = -y_{12}$ = **-5** = **-5.000** pu
8. $Y_{13} = Y_{31}$ (times −j): **-4** = **-4.000** pu
9. $Y_{23} = Y_{32}$ (times −j): **-2.5** = **-2.500** pu

Each entry is shown as a number to multiply by −j (a pure reactance gives y = −j/x). Example: x_{12} = 0.2 → Y_{11} = −j9 with x_{13} = 0.25.

Before you start: buses and the admittance matrix

A **bus** is just a junction (node) in the power network where generators, loads and lines meet - like a junction in a pipe network.

Admittance y = 1/z (z = impedance, in ohms or per unit); it says how easily current flows. Kirchhoff's current law (KCL) at each bus says: current injected into the bus = sum of currents leaving through the lines. Write KCL (Kirchhoff's current law) for every bus and you get a matrix equation **I = Y V**. Y_{bus} is the "wiring map" of the network in numbers.

Per unit (pu) = value divided by a base value, so 1.0 pu means rated.

Bus, types of buses and their significance (2023 5a, 2022 5a)

A **bus** is a node in the power system at which one or more lines, generators or loads are connected. For load flow, each bus has four quantities: real power P, reactive power Q, voltage magnitude $|V|$ and angle δ; two are specified and two are calculated. So there are **three types of buses**:

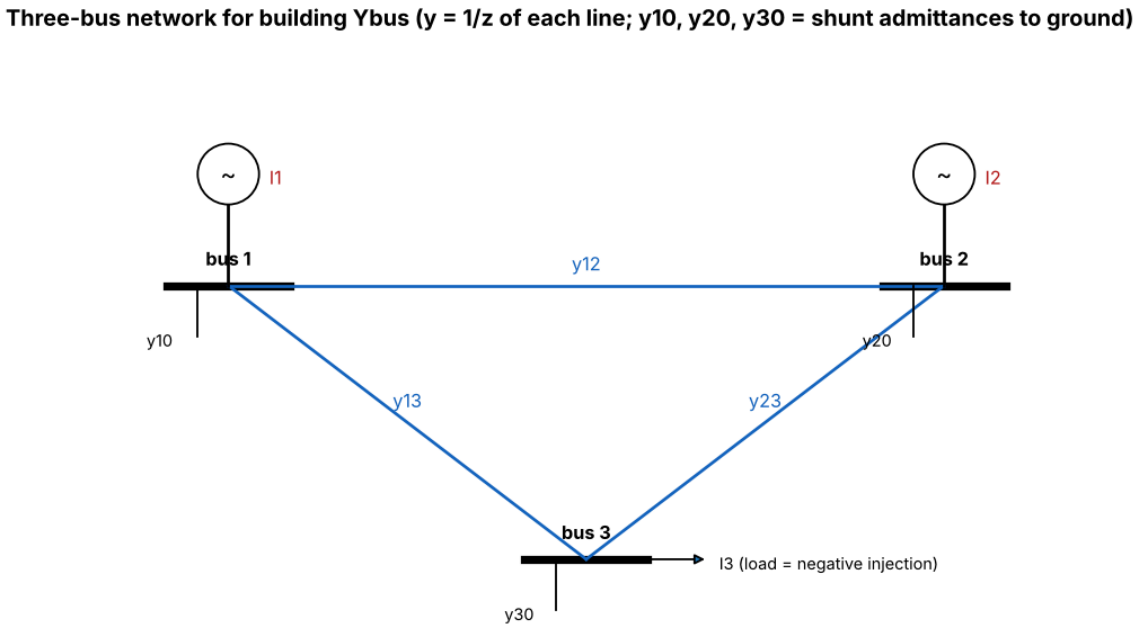
Bus type	Specified	Unknown	Significance
Slack (swing / reference) bus - only one	$\|V\|$ and δ (usually $1.0\angle 0°$)	P, Q	Provides the angle reference; supplies the system losses and balances generation and load, which are unknown before the solution
PV (generator / voltage-controlled) bus	P and $\|V\|$	Q, δ	Generators (and SVCs) whose excitation holds the voltage; Q must stay within limits (if a limit is hit, it becomes a PQ bus)
PQ (load) bus - most buses	P and Q	$\|V\|$, δ	Load buses (and buses with fixed-output DG (distributed generation)); their voltages are what the study checks

Classification is needed because the load-flow equations can be solved only when exactly two quantities per bus are known. In smart grids, DG with voltage control may be PV buses, DG at fixed power PQ buses, and droop-controlled units special buses.

Infinite bus (2023 5a)

An **infinite bus** is an idealised bus whose voltage magnitude and frequency remain constant regardless of how much power is injected or drawn - it has zero internal impedance and infinite inertia. A single generator connected to a very large grid sees the grid as an infinite bus; it is used in stability studies (B4) and in machine analysis (B5).

Deriving Y_{bus} and proving $I_{bus} = Y_{bus}V_{bus}$ (2023 5a)



Three-bus network used for the derivation (drawn for this pack)

Method: represent each line by its series admittance $y_{ij} = 1/z_{ij}$ and shunt admittances y_{i0} to ground; replace generators and loads by current injections I_i (positive into the bus). Apply KCL at each bus.

Step (write this)	Why / what it means
Bus 1: $I_1 = y_{10}V_1 + y_{12}(V_1 - V_2) + y_{13}(V_1 - V_3)$	Injected current = current to ground + currents leaving through each line (KCL). Line current = admittance × voltage difference.
$I_1 = (y_{10} + y_{12} + y_{13})V_1 - y_{12}V_2 - y_{13}V_3$	Collect the terms of each voltage.
Bus 2: $I_2 = -y_{12}V_1 + (y_{20} + y_{12} + y_{23})V_2 - y_{23}V_3$	Same for bus 2.
Bus 3: $I_3 = -y_{13}V_1 - y_{23}V_2 + (y_{30} + y_{13} + y_{23})V_3$	Same for bus 3 (a load bus has a negative injection).
Define $Y_{ii} = y_{i0} + \Sigma_j\, y_{ij}$ (self/driving-point admittance) and $Y_{ij} = -y_{ij}$ (mutual/transfer admittance)	Diagonal = sum of all admittances connected to the bus; off-diagonal = negative of the admittance joining the two buses (0 if not connected).
$I_i = \Sigma_{j=1}^{n}\, Y_{ij}V_j, \quad i = 1 \ldots n$	Every bus equation has this same form.
$[I_1; I_2; I_3] = [Y_{11}\ Y_{12}\ Y_{13};\ Y_{21}\ Y_{22}\ Y_{23};\ Y_{31}\ Y_{32}\ Y_{33}][V_1; V_2; V_3] \Rightarrow \mathbf{I_{bus} = Y_{bus}V_{bus}}$	Write the n equations in matrix form: proved.

Properties of Y_{bus}: square (n × n), symmetric ($Y_{ij} = Y_{ji}$ without phase shifters), complex, highly **sparse** (most buses are not directly connected), diagonally dominant; built easily by inspection; its inverse is Z_{bus}. It is the basis of Gauss-Seidel and Newton-Raphson load flow (A9) and of state estimation in the smart grid.

> **Memory trick: "Diagonal = sum of all, off-diagonal = minus the link"**. Bus types: **Slack (Vδ), PV, PQ** - the two letters are what you KNOW.

B4 Rotor angle stability: steady-state vs transient, swing equation, small-disturbance solution $\omega_n = \sqrt{(\pi f_0 P_s/H)}$, equal area criterion Must study

Very likely · 83% Hard

PAST-PAPER QUESTIONS

- **2023 Q6(a)** [12] In power system transient stability analysis, what principle does the equal area criterion (EAC) assess? Describe with necessary schematic diagram.
- **2022 Q6(a)** [8] Distinguish between transient and steady-state stability and discuss the need for performing stability analyses of power system.
- **2022 Q6(b)** [6] Elucidate the importance and role of the swing equation in power system stability analysis.
- **2022 Q6(c)** [14] Discuss the possible solutions of the disturbed motion of a synchronous generator connected to an infinite bus, when subjected to a small disturbance represented by $\Delta\delta$, and prove that $\omega_n = \sqrt{(\pi f_0 P_s/H)}$.

LEARN IT FROM ZERO

Rotor angle stability, from zero

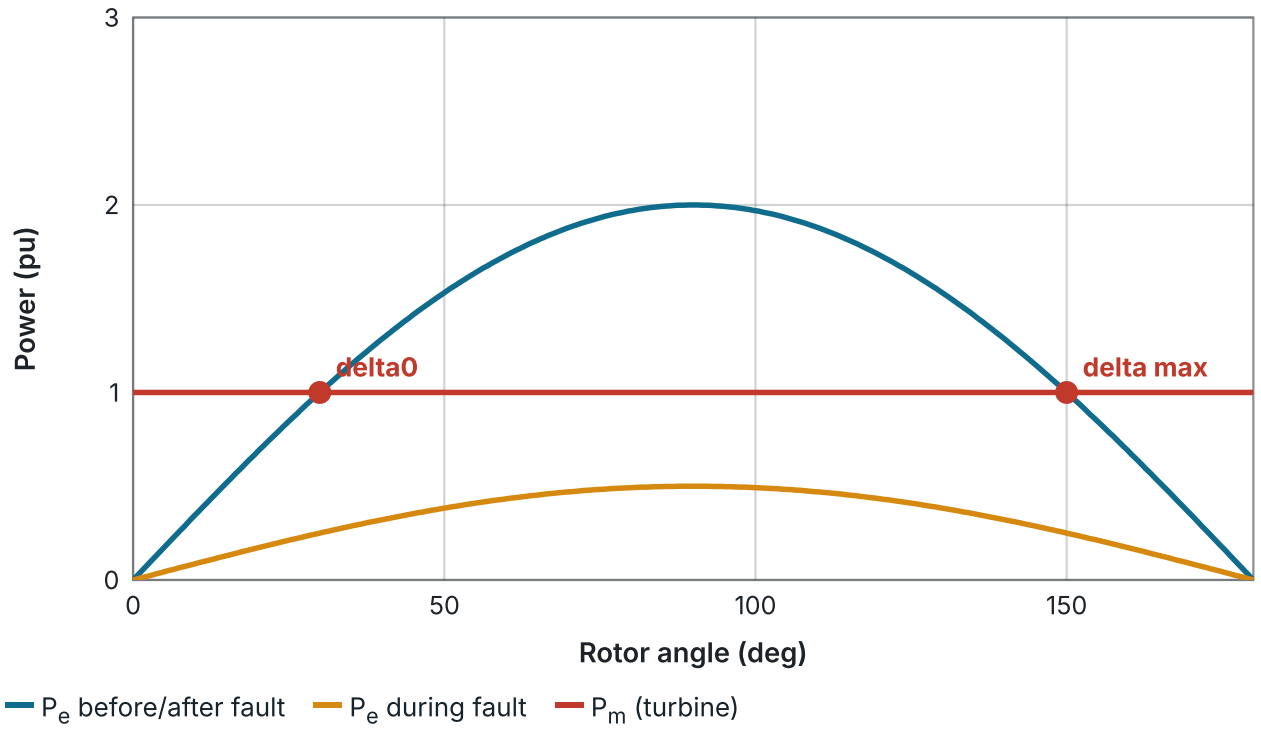
A generator sends power $P = (EV/X) \sin\delta$ to the grid, where δ is the angle between its internal voltage and the grid voltage. The turbine pushes with P_m; the grid pulls back with P_e. Where the two are equal (δ_0), the rotor sits steady.

After a fault P_e drops, so $P_m > P_e$ and the rotor **speeds up**: δ grows (area A_1, acceleration). When the fault is cleared P_e jumps above P_m and the rotor **slows down** (area A_2, deceleration). **Equal area criterion**: it is stable if the decelerating area available above P_m can be at least as large as the accelerating area. Clear the fault too late and A_1 becomes too big.

For a small push, the swing equation $(H/\pi f_0)\, d^2\Delta\delta/dt^2 + P_s\Delta\delta = 0$ is a spring-mass equation, so δ oscillates at $\boldsymbol{\omega_n = \sqrt{(\pi f_0 P_s/H)}}$, with $P_s = P_{max} \cos\delta_0$ (synchronising power coefficient). Stable only if $P_s > 0$, i.e. $\delta_0 < 90°$.

INTERACTIVE **Power-angle curve and equal areas**

Move the sliders and watch the curve change.



δ_0: **30.0 °** · δ_{max}: **150.0 °** · Critical clearing angle (fault P_e as set): **92.7 °**

Equal area: accelerate between δ_0 and the clearing angle (P_m above the fault curve), decelerate after it (P_e above P_m) up to δ_{max}.

CALCULATOR **Natural frequency of rotor oscillation (small disturbance)**

Change any number; every step is recalculated and shown with the numbers put in.

1. Operating angle: $\delta_0 = \sin^{-1}(P_0/P_{max}) = \sin^{-1}(\mathbf{1} / \mathbf{2}) =$ **30.00 °**
2. Synchronising power coefficient: $P_s = P_{max} \cos \delta_0 = \mathbf{2} \times \cos(\mathbf{30}) =$ **1.732** pu/rad
3. Natural frequency: $\omega_n = \sqrt{(\pi f_0 P_s / H)} = \sqrt{(\pi \times \mathbf{50} \times \mathbf{1.732} / \mathbf{5})} =$ **7.377** rad/s
4. In hertz: **7.377** / (2 × π) = **1.174** Hz
5. Period of swing: 1 / **1.174** = **0.85** s

Before you start: the swinging rotor

A synchronous generator connected to the grid is like a car towing a trailer with an elastic rope: the rotor angle δ is the stretch of the rope. Mechanical input power P_m pushes the rotor forward; electrical output $P_e = P_{max} \sin \delta$ pulls it back. If $P_m = P_e$, δ is steady.

After a disturbance, the rotor **swings** (oscillates) around a new angle. **Stable** = the swings die down; **unstable** = δ keeps growing and the machine loses synchronism (pole-slipping).

H = inertia constant (seconds) = stored kinetic energy / machine rating; f_0 = 50 Hz.

Transient vs steady-state stability and the need for stability analysis (2022 6a)

Steady-state (small-signal) stability	Transient stability
Ability to stay in synchronism after **small, gradual** disturbances (slow load changes)	Ability to stay in synchronism after a **large, sudden** disturbance (fault, loss of a line or generator, big load switching)
Equations can be linearised around the operating point	Non-linear equations; time-domain simulation or equal area criterion
Limit: steady-state stability limit (P_{max} = EV/X at $\delta = 90°$)	Limit: transient stability limit (lower than steady-state limit); critical clearing time
Depends on operating point, AVR (automatic voltage regulator), PSS (power system stabiliser)	Depends on fault type, location, clearing time, inertia, post-fault network

Need: to find maximum safe power transfers and operating limits; to set protection and critical fault clearing times; to design controllers (AVR, PSS, governors, FACTS (flexible AC transmission systems)); to plan new lines and generators; to prevent blackouts. In smart grids, inverter-based renewables reduce inertia and change dynamics, so stability analysis is even more important.

Swing equation: importance and role (2022 6b)

$(2H/\omega_s)\ d^2\delta/dt^2 = P_m - P_e$ or $(H/\pi f_0)\ d^2\delta/dt^2 = P_m - P_{max}\sin\delta$ (pu, δ in electrical radians)

It is Newton's second law for the rotor (inertia × angular acceleration = accelerating power). **Role:** it is the fundamental equation of rotor-angle stability - it describes how δ changes with time after a disturbance; solving it (numerically, step by step) shows whether δ settles (stable) or increases without limit (unstable); it gives the basis for the equal area criterion, small-signal analysis, critical clearing angle and time, and for designing controls and protection.

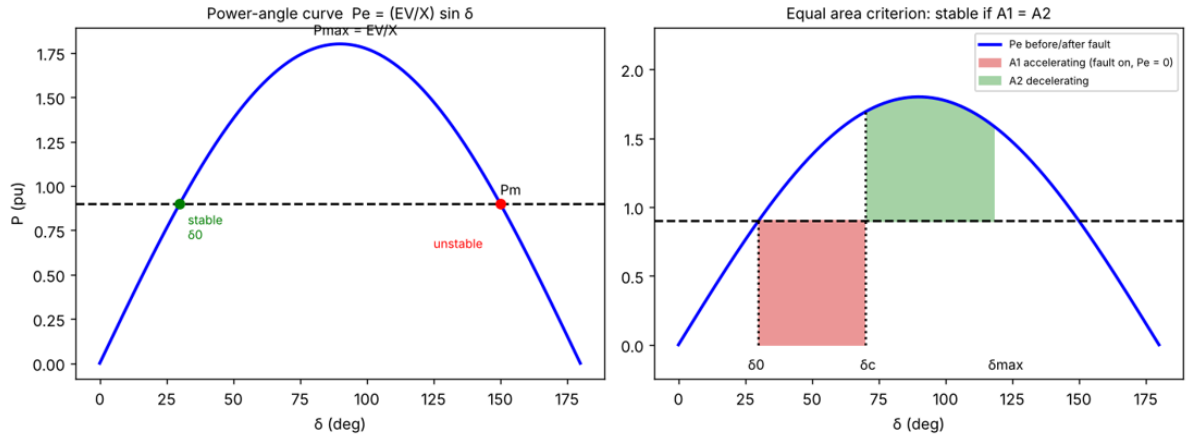
Solution for a small disturbance and proof of $\omega_n = \sqrt{(\pi f_0 P_s/H)}$ (2022 6c)

Step (write this)	Why / what it means
$(H/\pi f_0)\ d^2\delta/dt^2 + P_{max}\sin\delta = P_m$	Swing equation with $P_e = P_{max}\sin\delta$ (machine on an infinite bus).
Let $\delta = \delta_0 + \Delta\delta$; $\sin(\delta_0 + \Delta\delta) \approx \sin\delta_0 + \cos\delta_0 \cdot \Delta\delta$	Small disturbance: $\sin\Delta\delta \approx \Delta\delta$, $\cos\Delta\delta \approx 1$.
$(H/\pi f_0)\ d^2\Delta\delta/dt^2 + P_{max}\cos\delta_0 \cdot \Delta\delta = 0$	Initial balance $P_m = P_{max}\sin\delta_0$ cancels; $d^2\delta_0/dt^2 = 0$.
$P_s = dP_e/d\delta\ |_{\delta 0} = P_{max}\cos\delta_0$	Synchronising power coefficient = slope of the power-angle curve.
$(H/\pi f_0)\ d^2\Delta\delta/dt^2 + P_s\Delta\delta = 0$	A linear 2nd-order equation, like a mass on a spring.
Characteristic equation: $s^2 + (\pi f_0 P_s/H) = 0 \Rightarrow s = \pm j\sqrt{(\pi f_0 P_s/H)}$	Put $\Delta\delta = e^{st}$.
$\omega_n = \sqrt{(\pi f_0 P_s/H)}$ rad/s; $f_n = \omega_n/2\pi$	Proved: the natural frequency of rotor oscillation.
If $P_s > 0$ ($\delta_0 < 90°$): $\Delta\delta(t) = \Delta\delta_0 \cos\omega_n t$ → sustained oscillation, **stable**	Roots purely imaginary. With damping D the oscillations decay: $\Delta\delta = (\Delta\delta_0/\sqrt{(1-\zeta^2)})\ e^{-\zeta\omega n t}\sin(\omega_d t + \theta)$.

If $P_s < 0$ ($\delta_0 > 90°$): $s = \pm\sqrt{(\pi f_0 \lvert P_s \rvert / H)}$, one root positive → $\Delta\delta$ grows exponentially, **unstable**	Operating beyond the peak of the P-δ curve.

Side note: Typical numbers: H = 5 s, f_0 = 50 Hz, P_s = 1.5 pu → $\omega_n = \sqrt{(\pi \times 50 \times 1.5 / 5)} = 6.86$ rad/s, $f_n \approx 1.1$ Hz: the typical 1-2 Hz local mode of power swings.

Equal area criterion (2023 6a)



Power-angle curve and equal area criterion (drawn for this pack)

The **equal area criterion (EAC)** is a graphical method to judge the **transient stability** of a single machine connected to an infinite bus (or two machines) without solving the swing equation in time. It assesses whether the kinetic energy the rotor gains during a disturbance can be given back to the system before the angle passes the unstable point.

1. Before the fault the machine runs at δ_0 where $P_m = P_e$.
2. During a fault P_e drops (to zero for a three-phase fault at the terminals) while P_m is constant, so the rotor accelerates: kinetic energy gained = area $\mathbf{A_1} = \int(P_m - P_e)d\delta$ from δ_0 to the clearing angle δ_c.
3. After clearing, $P_e > P_m$, the rotor decelerates; it gives back energy = area $\mathbf{A_2} = \int(P_e - P_m)d\delta$ from δ_c to δ_{max}.
4. The rotor stops swinging forward when $A_2 = A_1$. The system is **stable if an A_2 equal to A_1 is available before δ reaches the unstable point $\delta_{max} = 180° - \delta_0$**.
5. The **critical clearing angle** δ_{cr} makes A_1 = maximum available A_2; for $P_e = 0$ during the fault: $\cos\delta_{cr} = (\pi - 2\delta_0)\sin\delta_0 - \cos\delta_0$; critical clearing time $t_{cr} = \sqrt{[4H(\delta_{cr} - \delta_0)/(\omega_s P_m)]}$.

Memory trick: Swing equation = **"Inertia × acceleration = $P_m - P_e$"**. Small signal: **$\omega_n = \sqrt{(\pi f_0 P_s / H)}$** - "pi f P over H, root it". EAC (equal area criterion): **"energy gained (A1) must be paid back (A2)"**.

B5 Numerical: voltage behind transient reactance and transient power-angle equation (with and without saliency) Good to have Possible · 17% Hard

PAST-PAPER QUESTIONS

- **2020 Q4(b)** [10] A synchronous machine has x_d = 1.0, x_q = 0.6, x'_d = 0.3 pu and negligible armature resistance. It is connected directly to an infinite bus of 1.0 pu and delivers 0.5 pu real power at 0.8 pf lagging. Determine the voltage behind transient reactance and the transient power-angle equation (i) neglecting saliency, (ii) including saliency.

UNDERSTAND **Read this to understand; you do not need to write it**

A generator in a transient study is a voltage E′ behind its transient reactance x'_d. Find E′ by walking from the infinite bus through x'_d: $\mathbf{E' = V + jx'_d I}$. Then $P_e = (E'V/x'_d)\sin\delta$. A salient-pole machine adds a reluctance term $(V^2/2)(1/x_q - 1/x'_d)\sin 2\delta$.

Always check: put δ_0 back in your equation; it must give the original P (0.5 pu here).

Symbols used here

Symbol	Name and meaning	Unit
x_d, x_q	Direct- and quadrature-axis synchronous reactance	pu
x'_d	Transient direct-axis reactance	pu
V	Infinite bus voltage (reference, 1∠0°)	pu
I, φ	Armature current and power-factor angle (lagging = negative)	pu, deg
E'	Voltage behind transient reactance	pu
E_Q	Fictitious emf on the q-axis (gives the true rotor angle)	pu
I_d	d-axis current component	pu

CALCULATOR **Voltage behind transient reactance and power-angle equation**

Change any number; every step is recalculated and shown with the numbers put in.

1. Current magnitude: $|I| = P/(V\,pf)$ = **0.5** / (**1** × **0.8**) = **0.6250** pu
2. Power-factor angle: $\cos^{-1}$(**0.8**) = **36.87** °
3. I real part: **0.625** × cos(**36.87**) = **0.5000** pu
4. I imaginary part (lagging): -**0.625** × sin(**36.87**) = **-0.3750** pu
5. E′ real = V − $x'_d\, I_{imag}$: **1** - **0.3** × **(-0.375)** = **1.1125** pu
6. E′ imag = $x'_d\, I_{real}$: **0.3** × **0.5** = **0.1500** pu
7. $|E'|$ (no saliency): √(**1.113** × **1.113** + **0.15** × **0.15**) = **1.1226** pu
8. δ_0 (no saliency): $\tan^{-1}$(**0.15** / **1.113**) = **7.68** °
9. $P_{max} = E'V/x'_d$: **1.123** × **1** / **0.3** = **3.742** pu

10. $|E_Q| = |V + jx_q I|$: √(pow(**1** - **0.6** × **(-0.375)**,2) + pow(**0.6** × **0.5**,2)) = **1.2612** pu
11. True rotor angle δ_0: $\tan^{-1}$(**0.6** × **0.5** / (**1** - **0.6** × **(-0.375)**)) = **13.76** °
12. $I_d = |I| \sin(\delta_0 + \varphi)$: **0.625** × sin(**13.76** + **36.87**) = **0.4832** pu
13. $E'_q = |E_Q| - (x_q - x'_d)I_d$: **1.261** - (**0.6** - **0.3**) × **0.4832** = **1.1162** pu
14. sin δ coefficient (salient): **1.116** × **1** / **0.3** = **3.721** pu
15. sin 2δ coefficient (salient): **1** × **1** / 2 × (1 / **0.6** - 1 / **0.3**) = **-0.833** pu
16. Check P at δ_0: **3.721** × sin(**13.76**) + **(-0.8333)** × sin(2 × **13.76**) = **0.500** pu

Before you start: the machine behind a reactance

For transient studies a synchronous machine is drawn as a voltage source E′ behind its **transient reactance** x'_d. We find E′ from the operating point: E′ = V + j x I (phasor KVL (Kirchhoff's voltage law)).

A **salient-pole** machine (projecting poles, e.g. hydro) has different reactances along the pole axis (d-axis) and between poles (q-axis), so its power-angle curve gets an extra sin 2δ term (reluctance power).

j = √−1 rotates a phasor by 90°. Lagging pf 0.8 means current lags voltage by $\cos^{-1}0.8 = 36.87°$.

Numerical (2020 4b)

Data: $x_d = 1.0$, $x_q = 0.6$, $x'_d = 0.3$ pu, V = 1.0∠0° pu (infinite bus), P = 0.5 pu, pf = 0.8 lagging.

Step (write this)	Why / what it means
`|I| = P/(V·pf) = 0.5/(1 × 0.8) = 0.625 pu; φ = -36.87°`	Current magnitude from P = VI cosφ; lagging means negative angle.
`I = 0.625∠-36.87° = 0.5 - j0.375 pu`	Write in rectangular form (0.625 × 0.8 = 0.5, 0.625 × 0.6 = 0.375).

(i) Neglecting saliency (x'_d behind the bus):

Step (write this)	Why / what it means
`E′ = V + jx′dI = 1 + j0.3(0.5 - j0.375)`	KVL from the infinite bus to the internal emf.
`= 1 + 0.1125 + j0.15 = 1.1125 + j0.15`	j × (−j) = +1.
`E′ = 1.1226∠7.68° pu`	$\|E'\| = \sqrt{(1.1125^2 + 0.15^2)}$; $\delta_0 = \tan^{-1}(0.15/1.1125)$.
`Pe = (E′V/x′d) sinδ = (1.1226 × 1/0.3) sinδ = 3.742 sinδ`	Transient power-angle equation. Check: 3.742 sin7.68° = 0.50 pu ✓

(ii) Including saliency:

Step (write this)	Why / what it means
`EQ = V + jxqI = 1 + j0.6(0.5 - j0.375) = 1.225 + j0.3 = 1.2612∠13.76°`	This fictitious emf lies on the q-axis, so it gives the true rotor angle $\delta_0 = 13.76°$.
`Id = |I| sin(δ0 + |φ|) = 0.625 sin(13.76° + 36.87°) = 0.625 × 0.7731 = 0.4832 pu`	d-axis component of the current (angle between I and the q-axis is $\delta_0 + \|\varphi\|$).

$E'_q = \|E_Q\| - (x_q - x'_d)I_d = 1.2612 - (0.6 - 0.3)(0.4832)$	Voltage behind transient reactance on the q-axis.
$E'_q = 1.1162$ pu (at $\delta_0 = 13.76°$)	
$P_e = (E'_q V/x'_d) \sin\delta + (V^2/2)(1/x_q - 1/x'_d) \sin2\delta$	Transient power-angle equation of a salient-pole machine ($x'_q = x_q$).
$= (1.1162/0.3) \sin\delta + 0.5(1/0.6 - 1/0.3) \sin2\delta$	Substitute.
$P_e = 3.721 \sin\delta - 0.833 \sin2\delta$	Check at 13.76°: 3.721(0.2378) − 0.833(0.4621) = 0.885 − 0.385 = 0.50 pu ✓

Side note: Saliency shifts the rotor angle (13.76° vs 7.68°) and adds a negative reluctance term during the transient (because $x'_d < x_q$). Always check your equation gives back P = 0.5 at δ_0.

Memory trick: "Current first, then E = V + jxI, then check P". Salient: **x_q gives the angle, I_d corrects the emf**.

WRITE IN EXAM **Write this in the exam (this much is enough for full marks)**

2020 4(b) Given: $x_d = 1.0$, $x_q = 0.6$, $x'_d = 0.3$ pu, $V = 1\angle 0°$, P = 0.5 pu, pf 0.8 lagging

$|I| = 0.5/(1 \times 0.8) = 0.625$ pu; $I = 0.625\angle -36.87° = 0.5 - j0.375$ pu

(i) Neglecting saliency: $E' = V + jx'_dI$

$= 1 + j0.3(0.5 - j0.375) = 1.1125 + j0.15$

$=$ **$1.1226\angle 7.68°$ pu**

$P_e = (E'V/x'_d) \sin\delta =$ **$3.742 \sin\delta$**

(ii) With saliency: $E_Q = V + jx_qI$

$= 1 + j0.6(0.5 - j0.375) = 1.225 + j0.3$

$=$ **$1.2612\angle 13.76°$**

$I_d = 0.625 \sin(13.76° + 36.87°) = 0.4832$ pu; $E'_q = 1.2612 - (0.6 - 0.3)(0.4832) = 1.1162$ pu

$P_e = (1.1162/0.3) \sin\delta + 0.5(1/0.6 - 1/0.3) \sin2\delta =$ **$3.721 \sin\delta - 0.833 \sin2\delta$**

Check at $\delta = 13.76°$: 0.885 − 0.385 = 0.50 pu ✓